Elektrizität und Magnetismus

Kapitel 2:
Stromkreise

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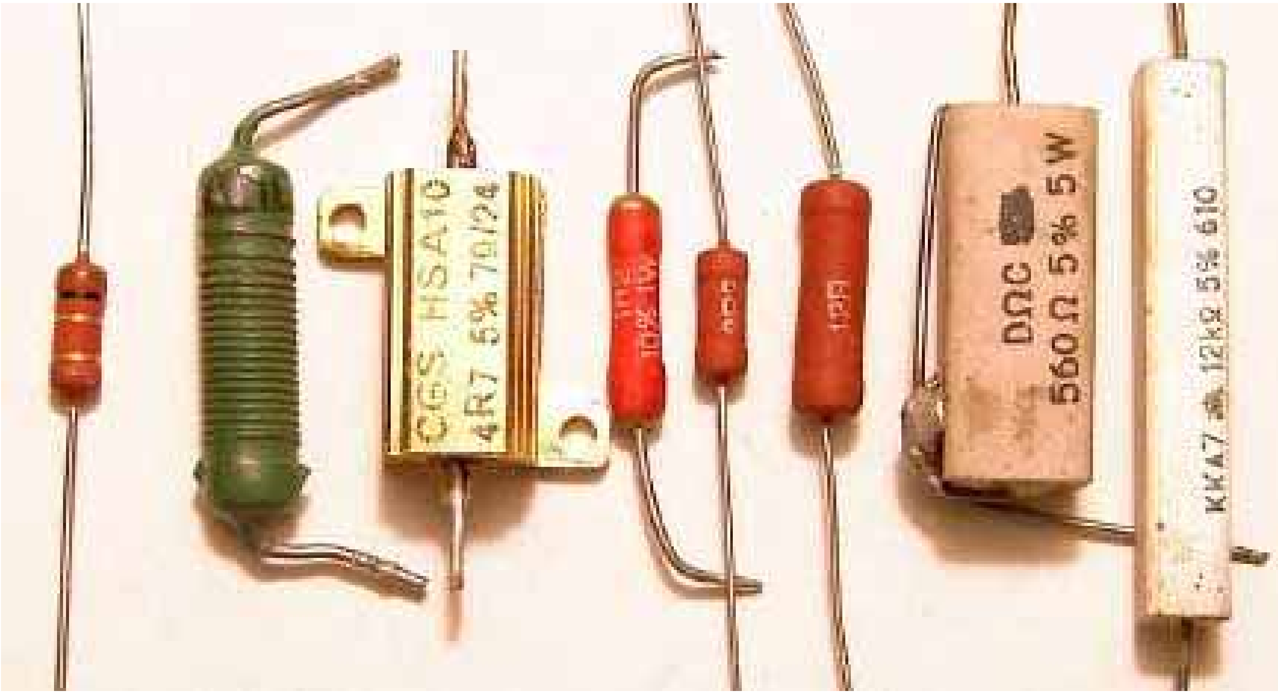
zu 2.1: Elektrischer Widerstand



schulphysikwiki.de

zu 2.1: Widerstandsmessung

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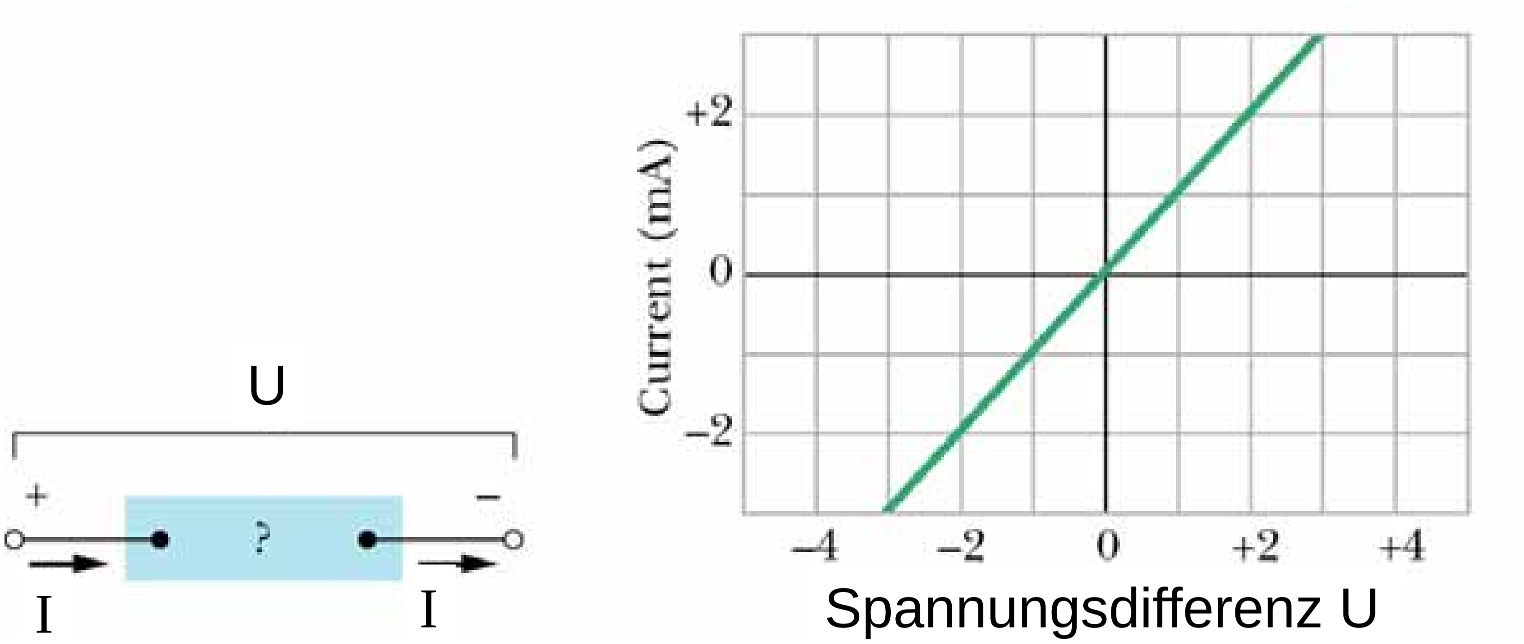
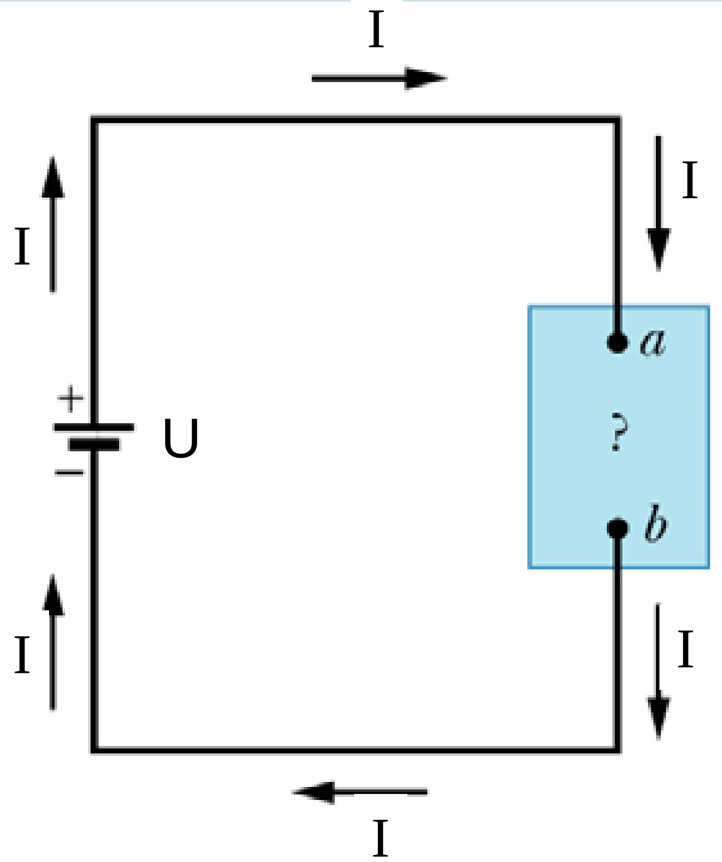


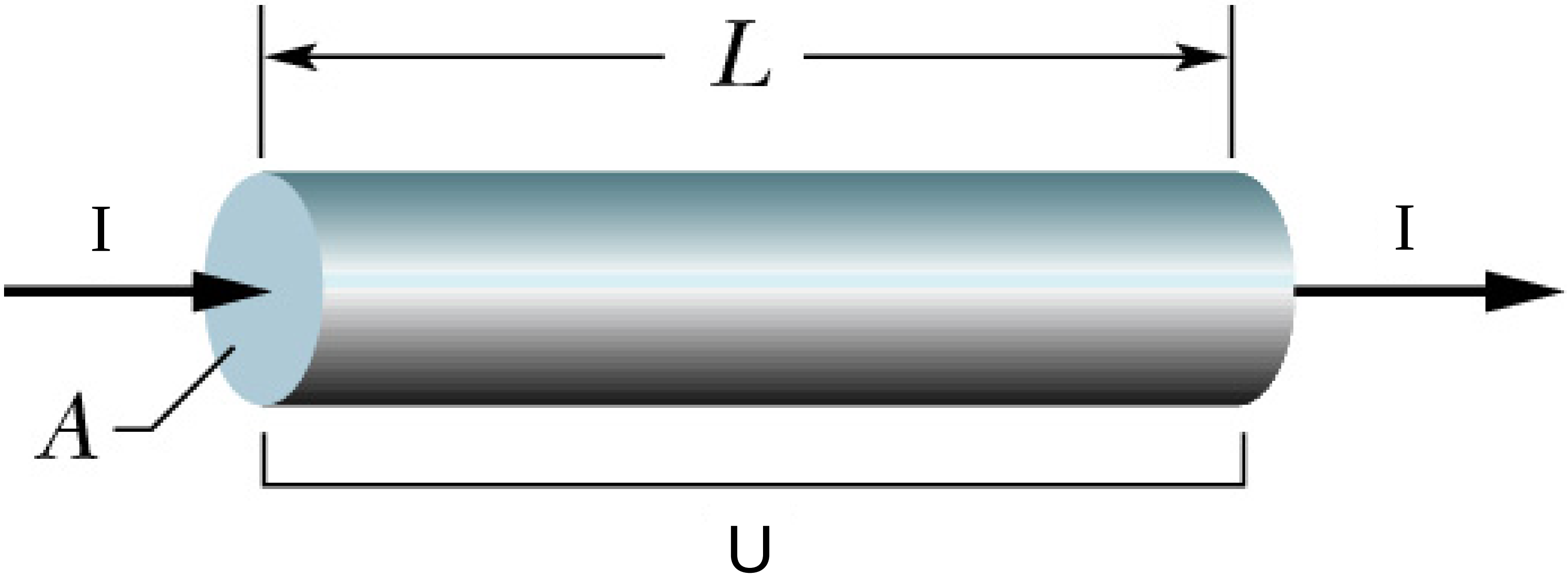
Halliday-Resnick-Walker (2003) Abb. 27.11

zu 2.1: Leistungsaufnahme eines Widerstands



Halliday-Resnick-Walker (2003) Abb. 27.13

zu - 2.1: Spezifischer Widerstand



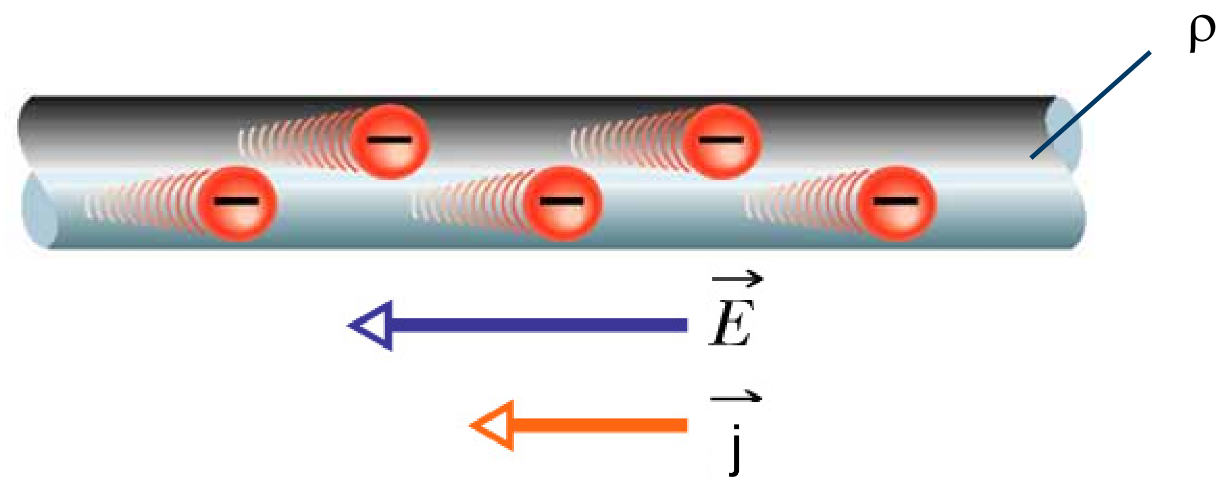
Halliday-Resnick-Walker (2003) Abb. 27.9

zu 2.1: Spezifischer Widerstand

Stoff	Spezifischer Widerstand ρ ($\Omega \cdot m$)
	Typische Metalle
Silber	$1{,}62 \cdot 10^{-8}$
Kupfer	$1{,}69 \cdot 10^{-8}$
Aluminium	$2{,}75 \cdot 10^{-8}$
Wolfram	$5{,}25 \cdot 10^{-8}$
Eisen	$9{,}68 \cdot 10^{-8}$
Platin	$10{,}6 \cdot 10^{-8}$
Manganin[a]	$4{,}82 \cdot 10^{-8}$

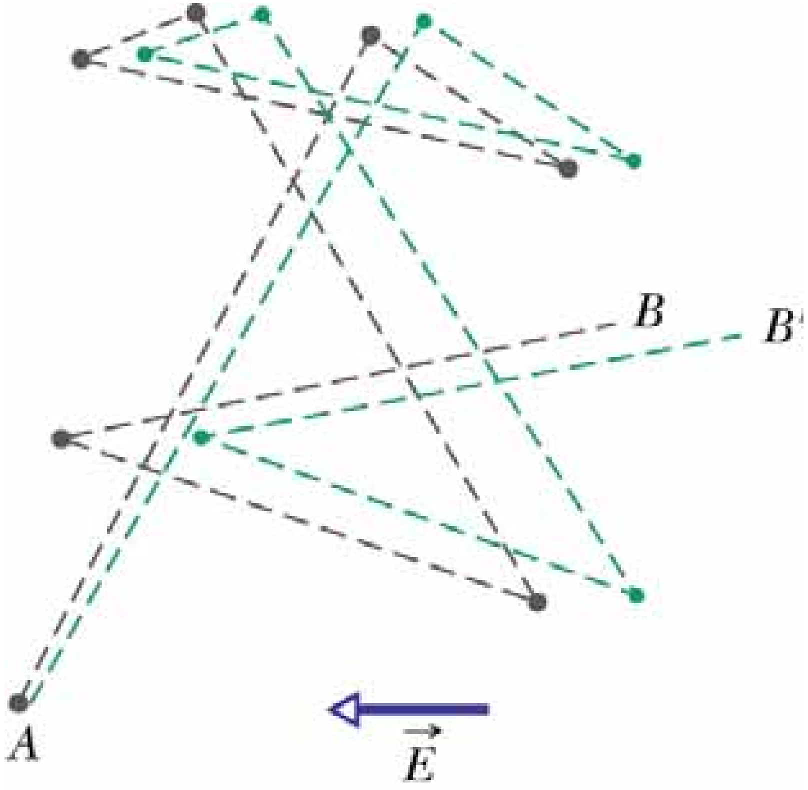
Halliday-Resnick-Walker (2003) Tab. 27.1

zu 2.1: Spezifischer Widerstand



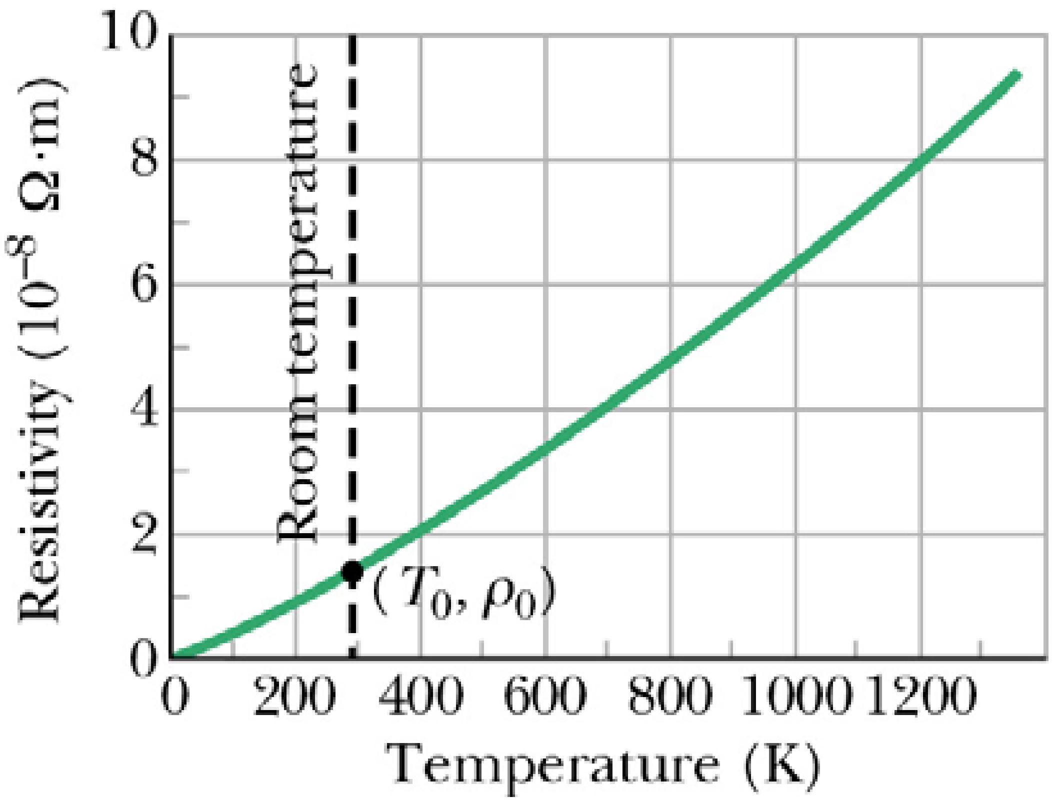
Halliday-Resnick-Walker (2003) Abb. 27.5

zu 2.1: Spezifischer Widerstand (zur Herleitung)



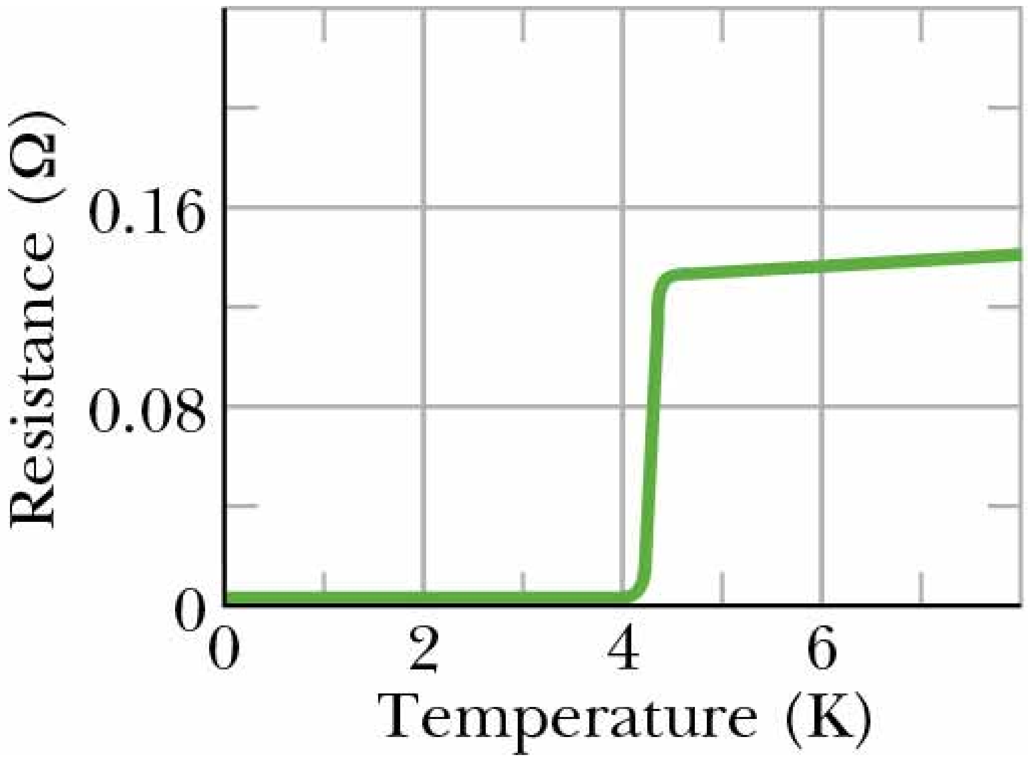
Halliday-Resnick-Walker (2003) Abb.. 27-12

zu 2.1: Temperaturabhängigkeit des Widerstands



Halliday-Resnick-Walker (2003) Abb. 27.9

zu 2.1: Widerstand eines Supraleiters



Halliday-Resnick-Walker (2003) Abb. 27.14

zu 2.1: Supraleiter als Stromtrasse von morgen?

www.mdr.de (24.4.2023)

zu 2.2: Batterien

wikipedia.org

electrive.net

zu 2.2: Solarkraftwerk

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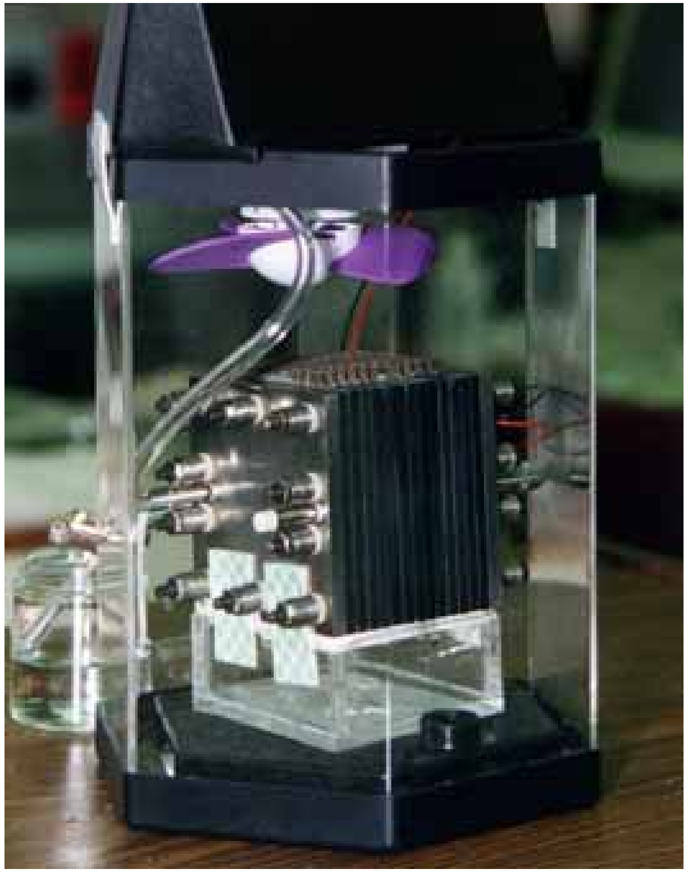
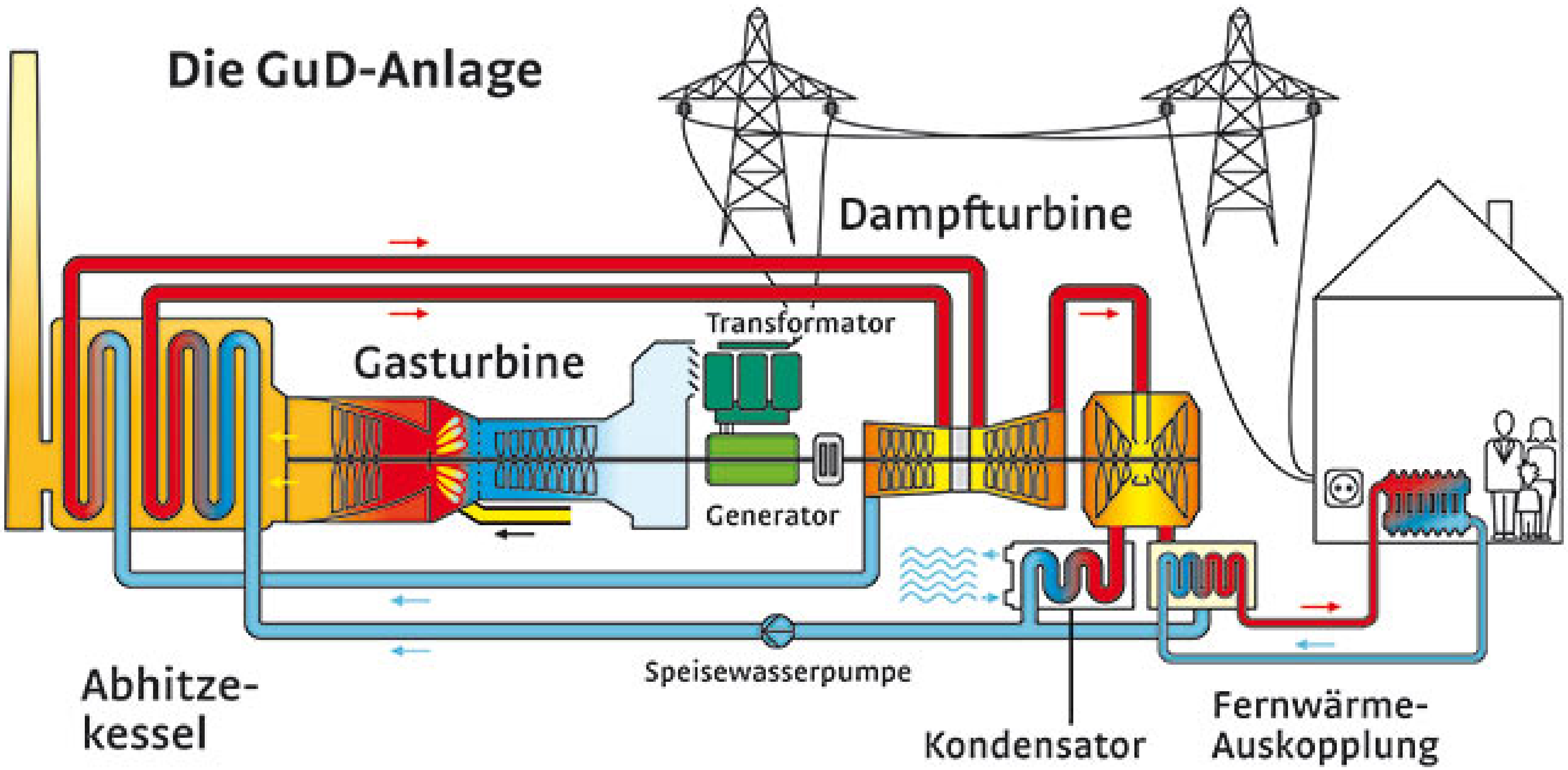
zu 2.2: Brennstoffzelle



wikipedia.org.

zu 2.2: Generator



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zu 2.2: Generator

Dampfturbine

Generator

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zu 2.2: Laborspannungsquelle

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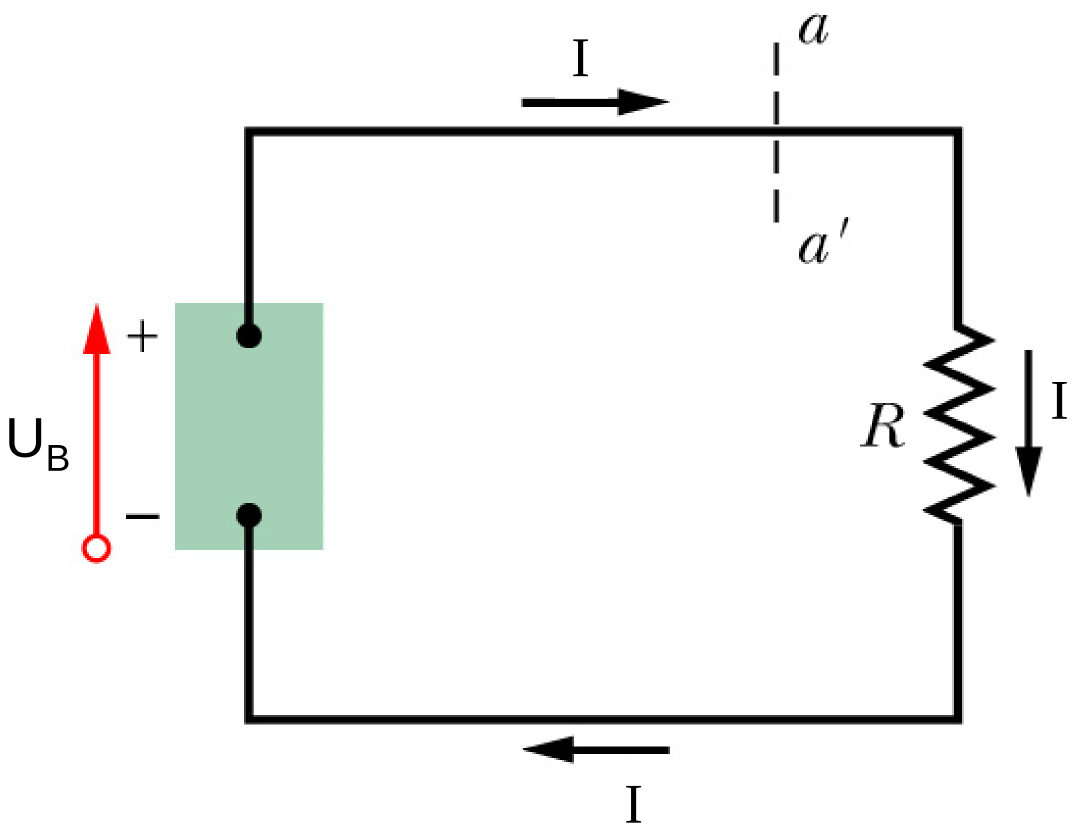
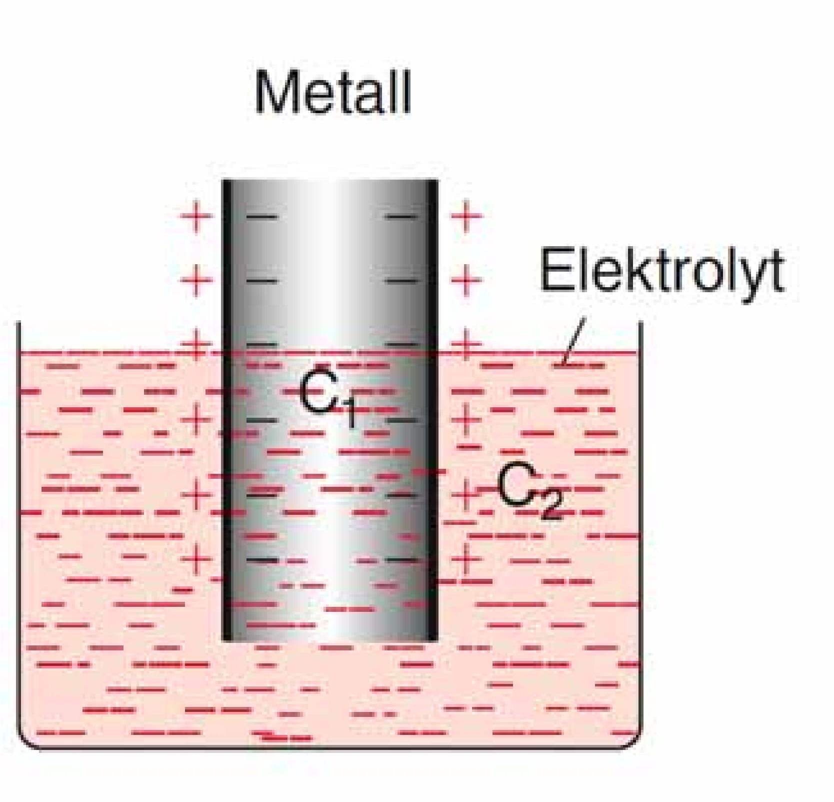
zu 2.2: Spannungsquelle mit Verbraucher



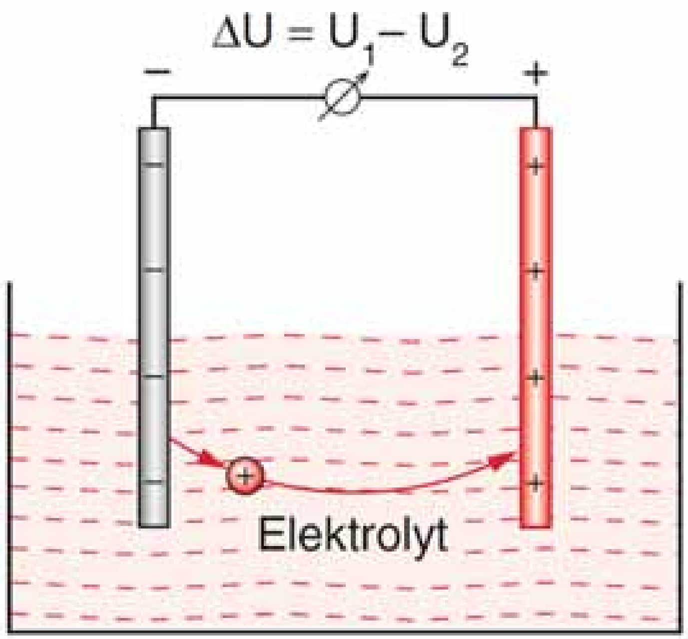
Halliday-Resnick-Walker (2003) Abb. 28.1

zu 2.2: Galvanische Zelle



Demtröder – Experimentalphysik 2 (2013), Abb. 2.46.

zu 2.2: Galvanische Zelle

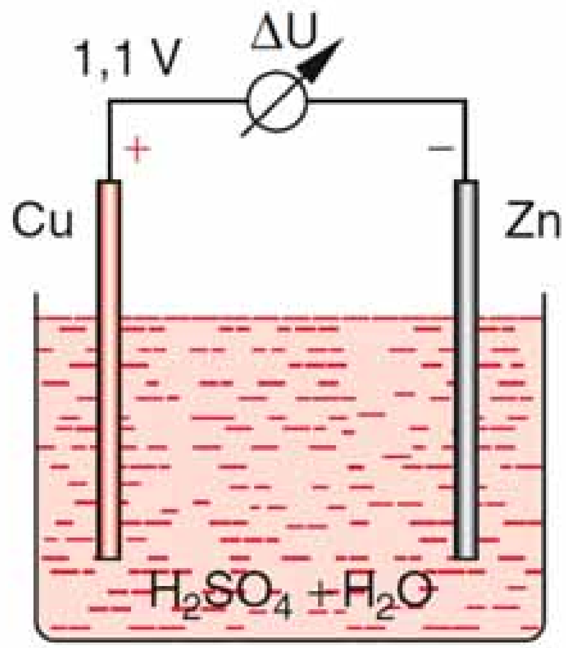


Demtröder – Experimentalphysik 2 (2013), Abb. 2.46.

Elektrode	U/V	Elektrode	U/V
Li^+/Li	−3,02	Ni^{++}/Ni	−0,25
K^+/K	−2,92	Pb^{++}/Pb	−0,126
Na^+/Na	−2,71	$H_2/2H^+$	0
Zn^{++}/Zn	−0,76	Cu^+/Cu	+0,35
Fe^{++}/Fe	−0,44	Ag^+/Ag	+0,8
Cd^{++}/Cd	−0,40	Au^{3+}/Au	+1,5

Demtröder – Experimentalphysik 2 (2013), Tab. 2.6.

zu 2.2: Galvanische Zelle



Demtröder – Experimentalphysik 2 (2013), Abb. 2.48.

Elektrode	U/V	Elektrode	U/V
Li^+/Li	$-3{,}02$	Ni^{++}/Ni	$-0{,}25$
K^+/K	$-2{,}92$	Pb^{++}/Pb	$-0{,}126$
Na^+/Na	$-2{,}71$	$H_2/2H^+$	0
Zn^{++}/Zn	$-0{,}76$	Cu^+/Cu	$+0{,}35$
Fe^{++}/Fe	$-0{,}44$	Ag^+/Ag	$+0{,}8$
Cd^{++}/Cd	$-0{,}40$	Au^{3+}/Au	$+1{,}5$

Demtröder – Experimentalphysik 2 (2013), Tab. 2.6.

zu 2.2: Galvanische Zelle

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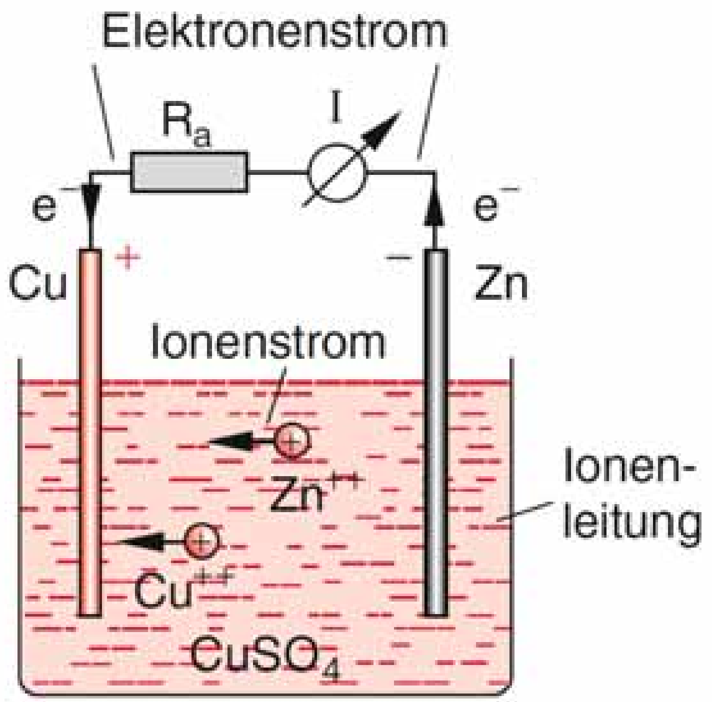
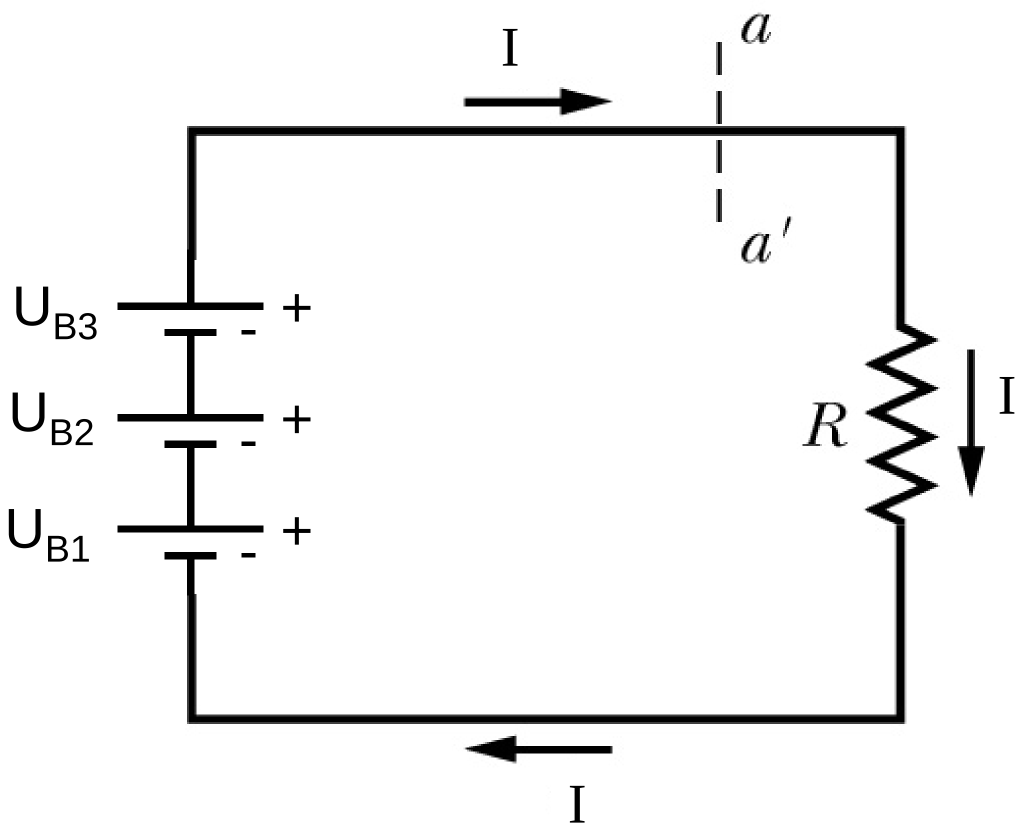


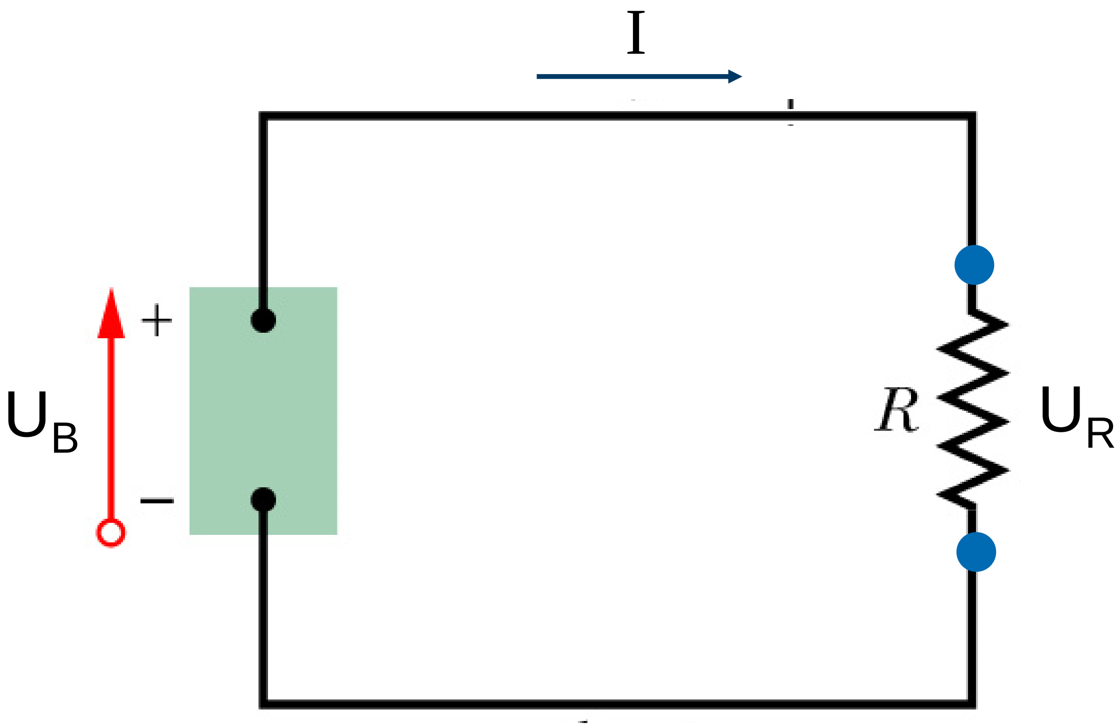
Demtröder – Experimentalphysik 2 (2013), Abb. 2.49.

zu 2.2: Reihenschaltung von Spannungsquellen



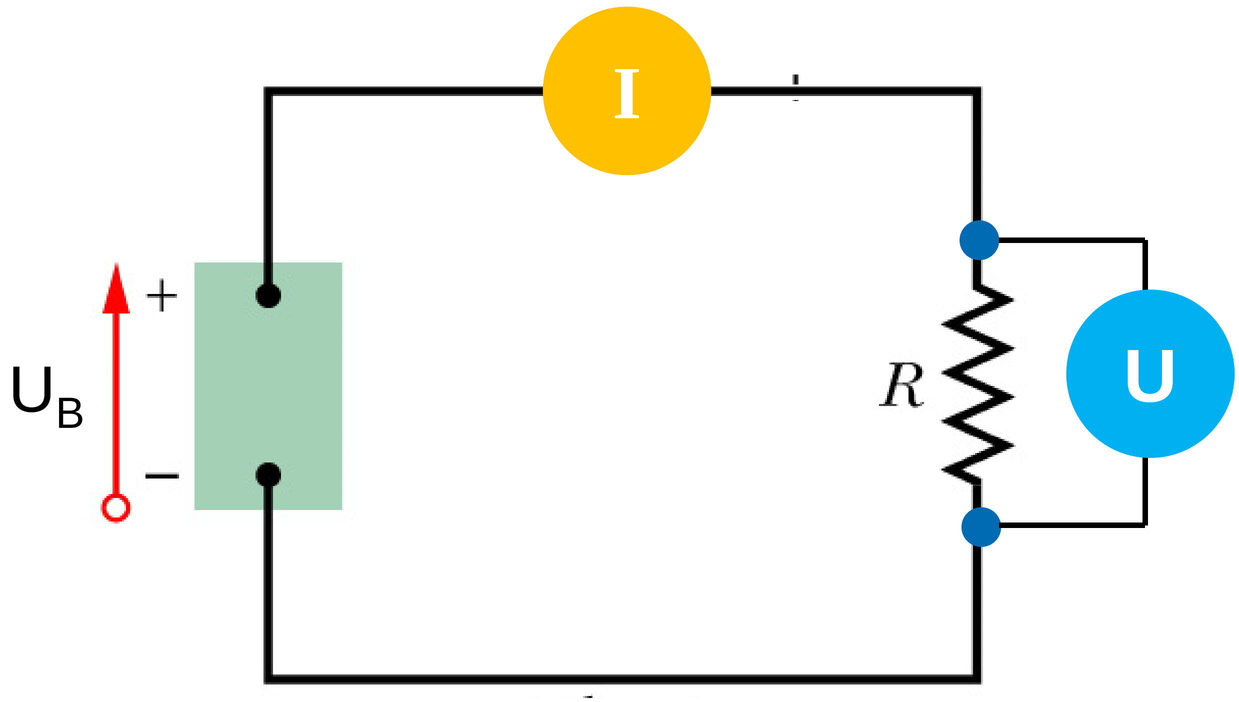
Halliday-Resnick-Walker (2003) Abb. 28.1

zu 2.3: Zur Maschenregel



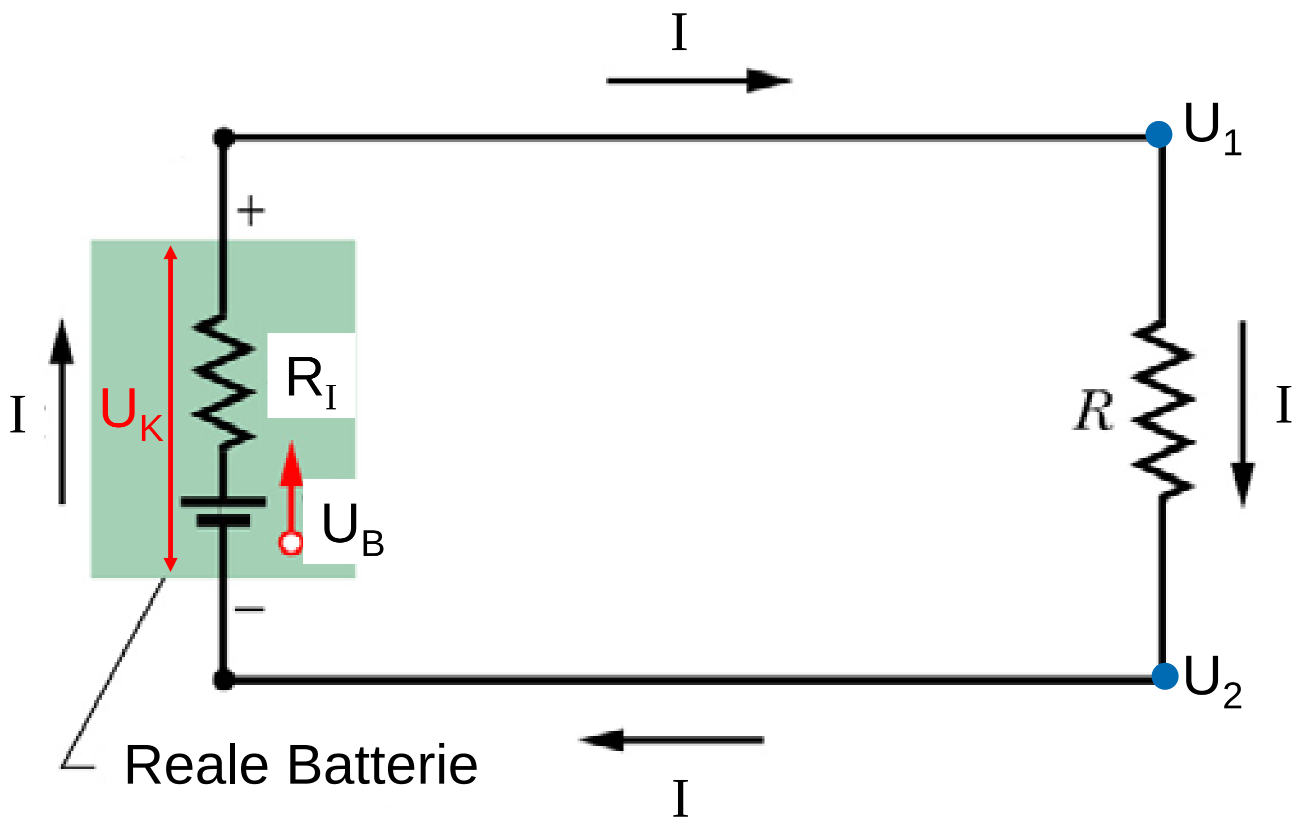
Halliday-Resnick-Walker (2003) Abb. 28.1

zu 2.3: Messung von Strom und Spannung



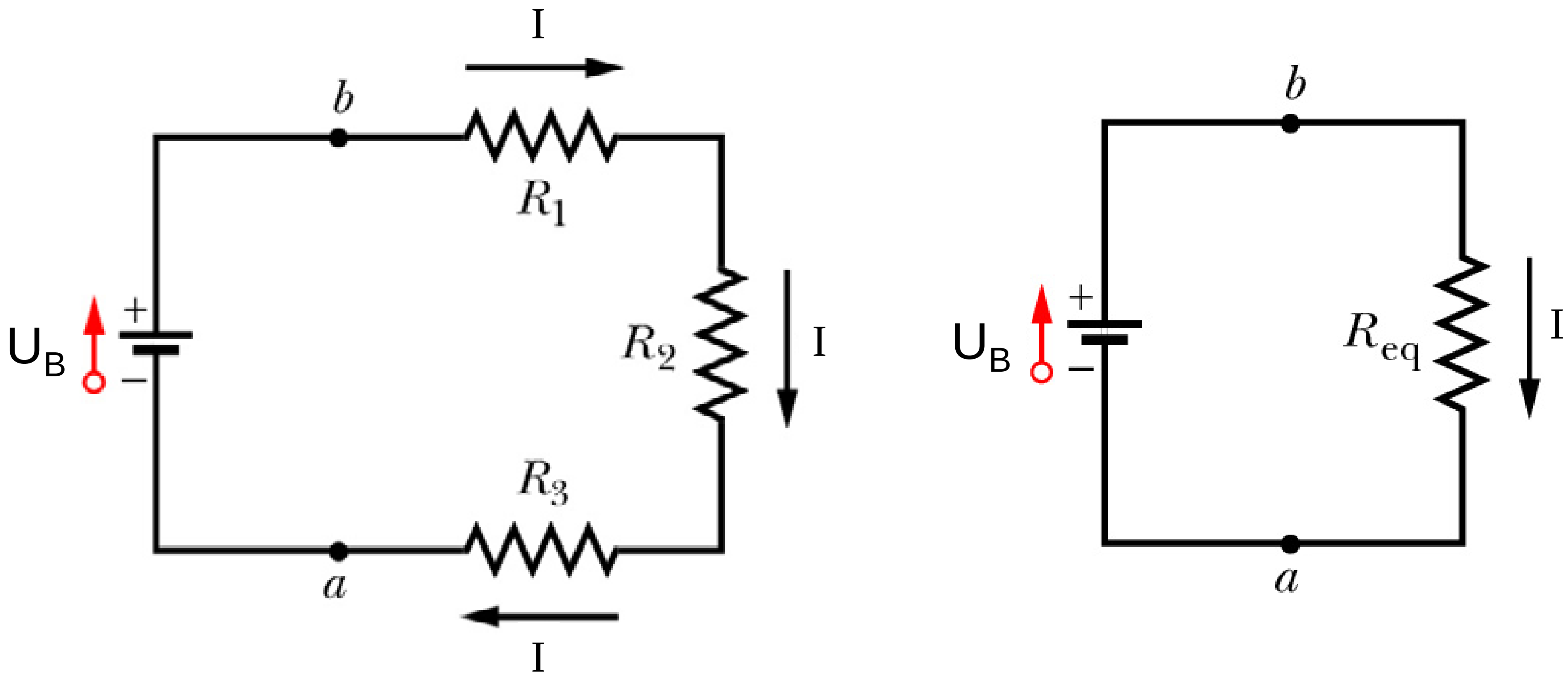
Halliday-Resnick-Walker (2003) Abb. 28.1

zu 2.3: Reale Spannungsquelle



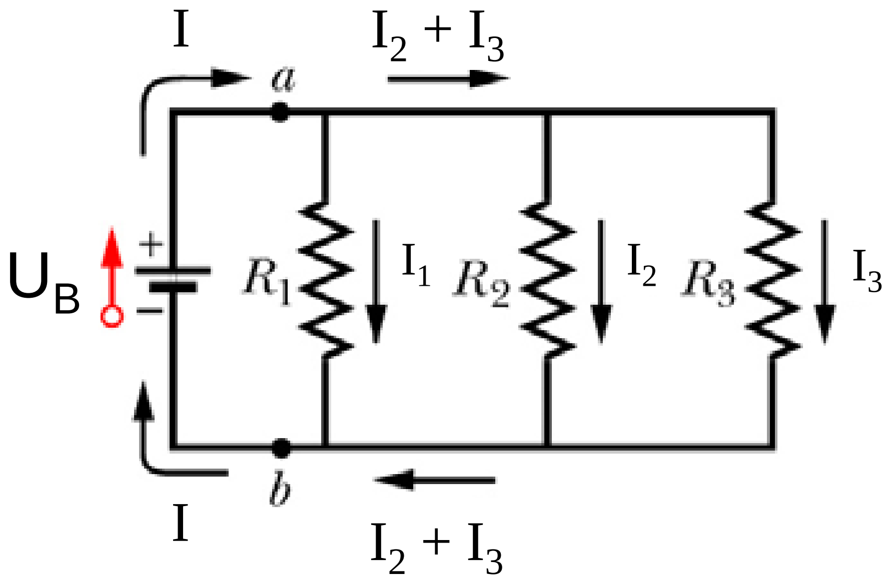
Halliday-Resnick-Walker (2003) Abb. 28.4

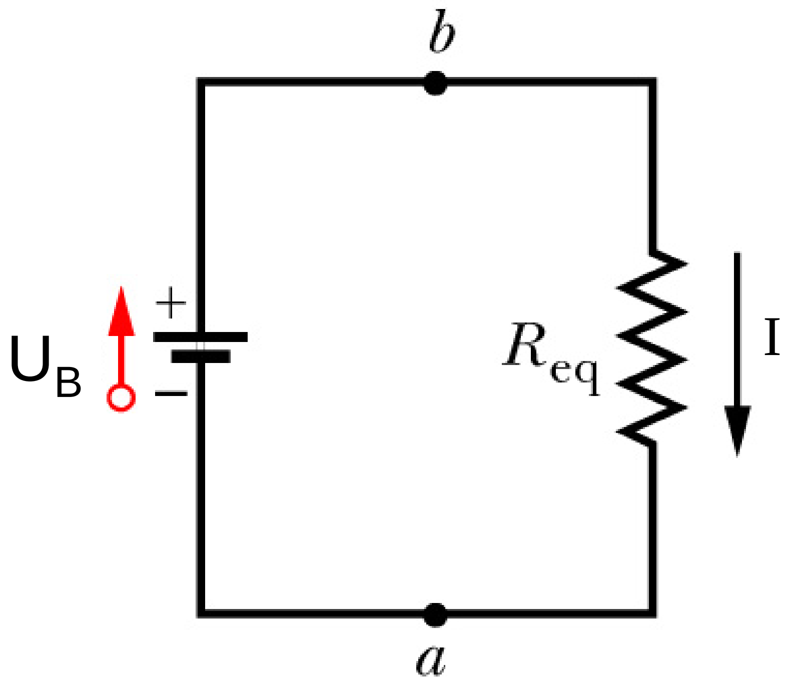
zu 2.3: Reihenschaltung von Widerständen



Halliday-Resnick-Walker (2003) Abb. 28.5

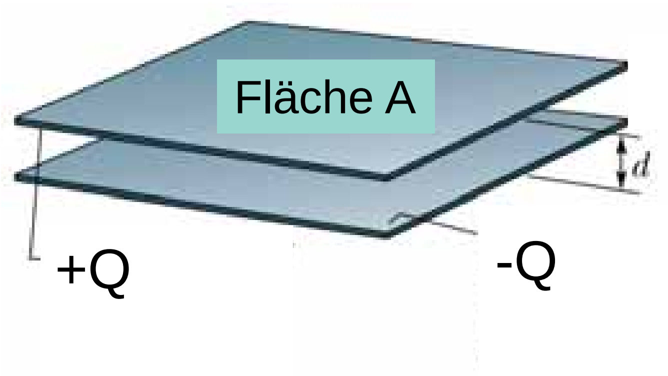
zu 2.3: Parallelschaltung von Widerständen





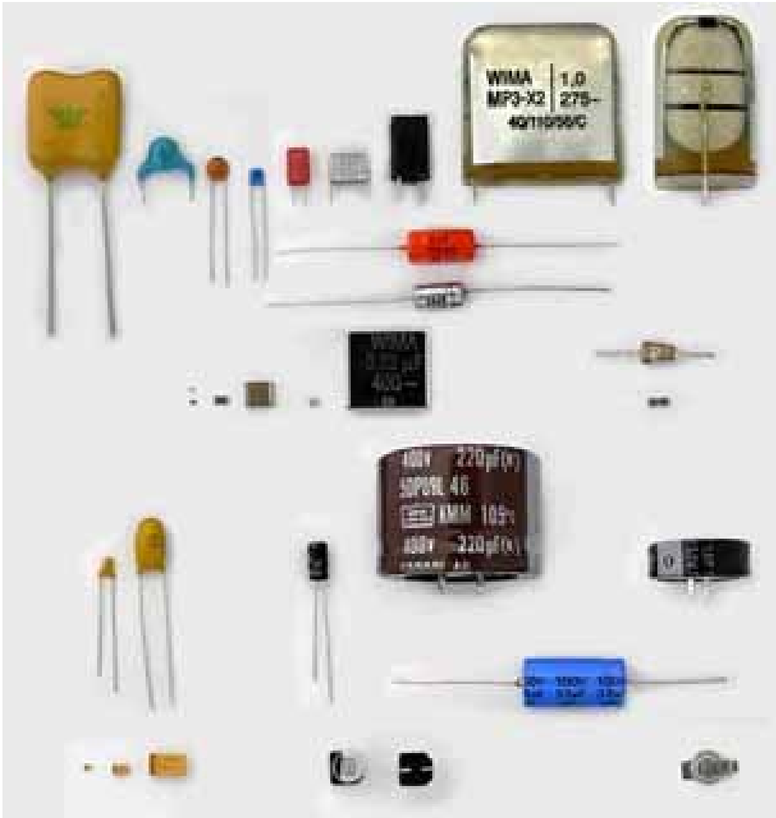
Halliday-Resnick-Walker (2003) Abb. 28.8

zu 2.4: Der Plattenkondensator



Halliday-Resnick-Walker (2003) Abb. 26.3

zu 2.4: Kondensatoren



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zu 2.4: Laden eines Plattenkondensators

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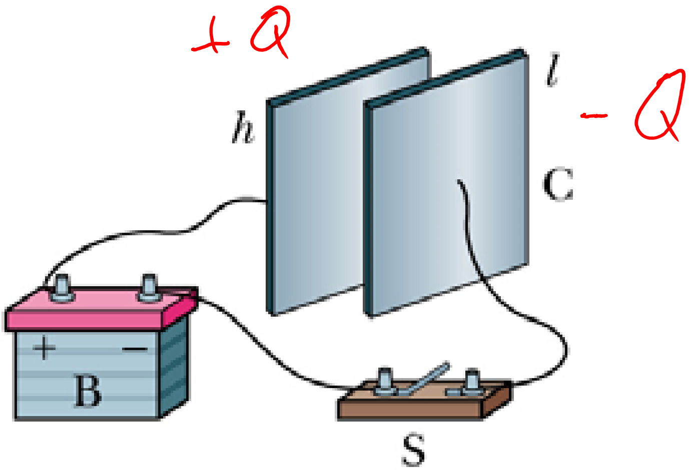
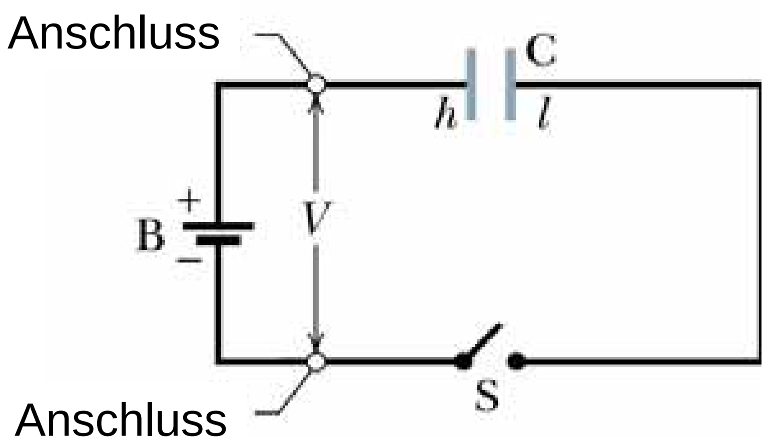
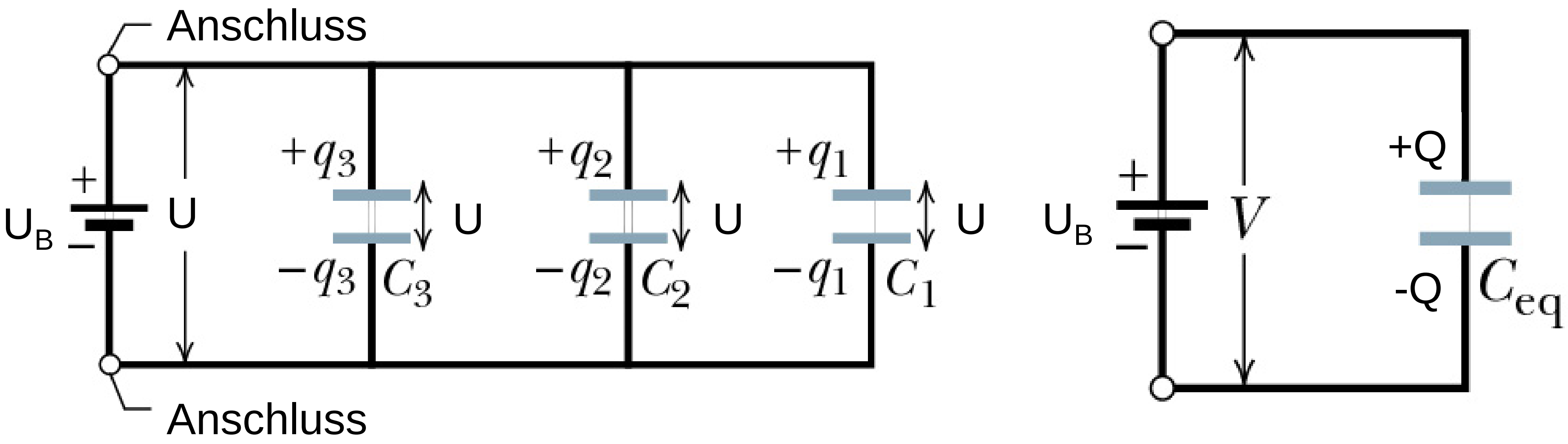




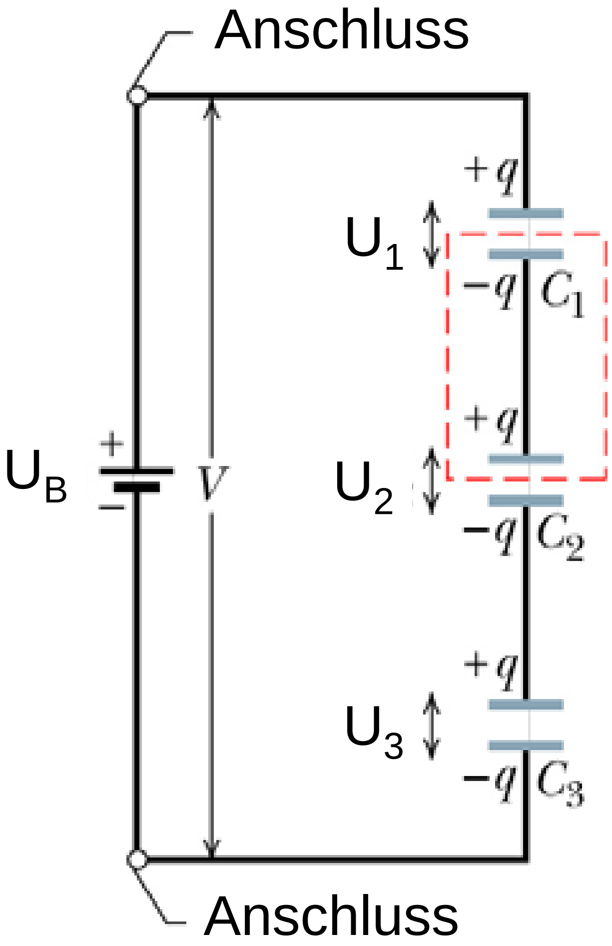
Halliday-Resnick-Walker (2003) Abb. 26.4

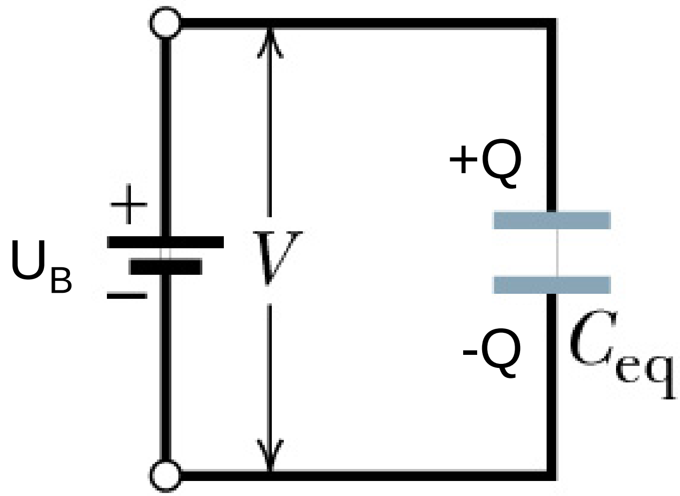
zu 2.4: Parallelschaltung von Kondensatoren



Halliday-Resnick-Walker (2003) Abb. 26.7

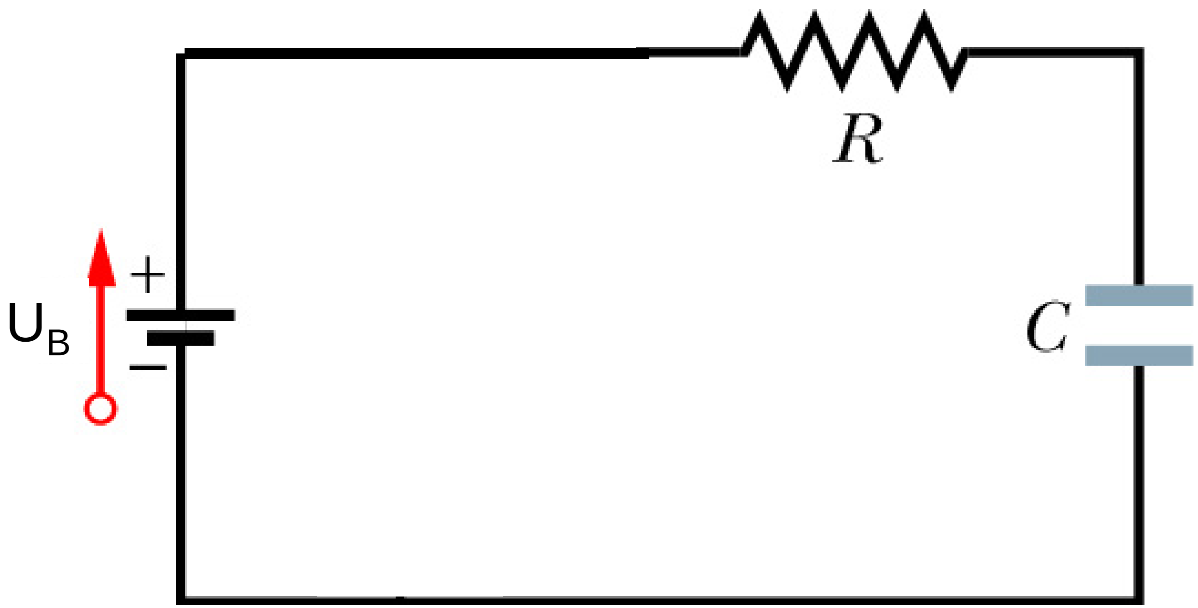
zu 2.4: Reihenschaltung von Kondensatoren





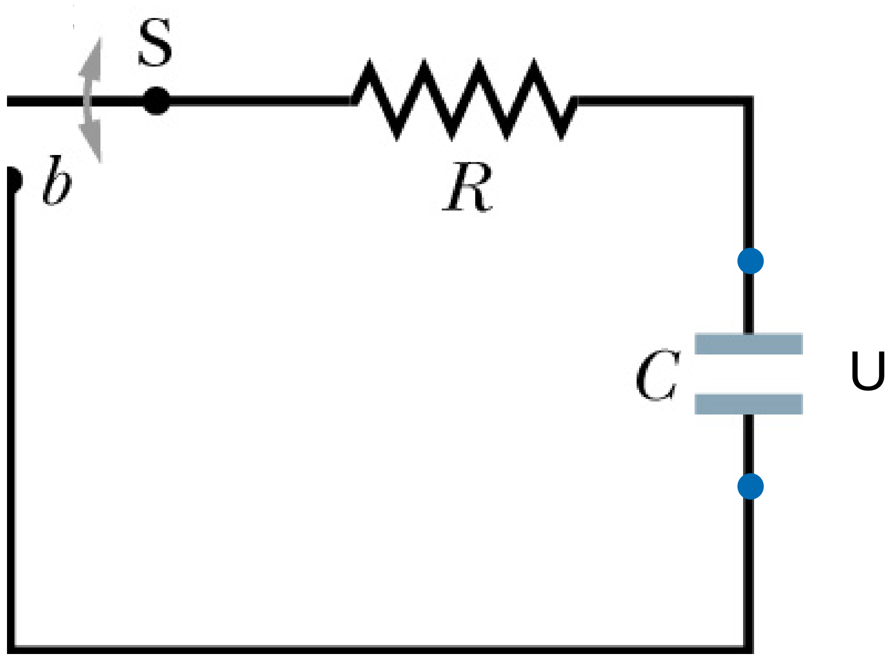
Halliday-Resnick-Walker (2003) Abb. 26.8

zu 2.4: Der RC-Kreis



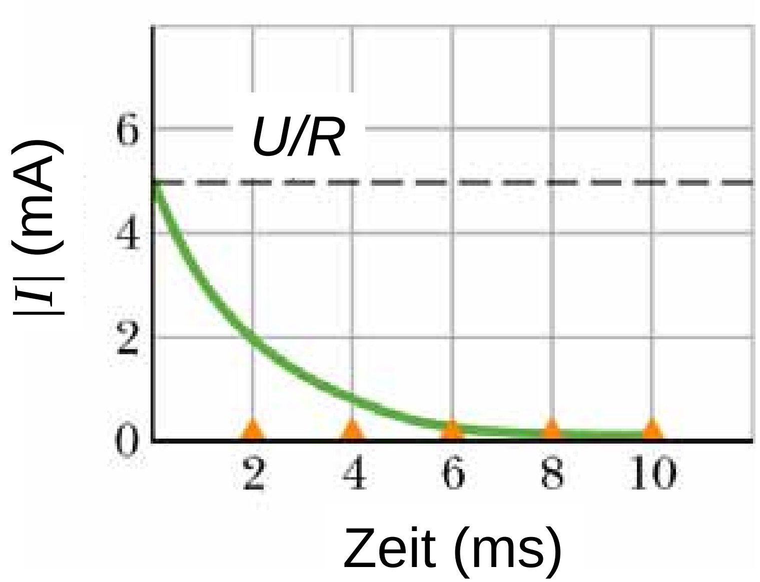
Halliday-Resnick-Walker (2003) Abb. 26.13

zu 2.4: Entladen eines Kondensators



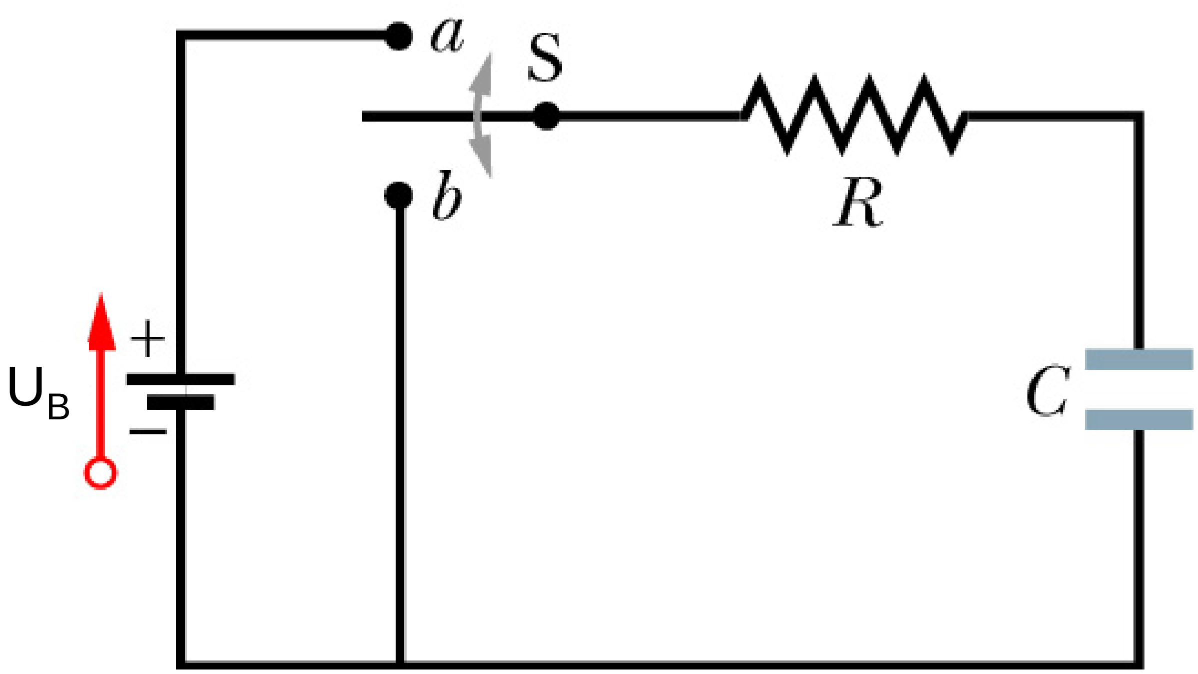
Halliday-Resnick-Walker (2003) Abb. 26.13

zu 2.4: Entladen eines Kondensators



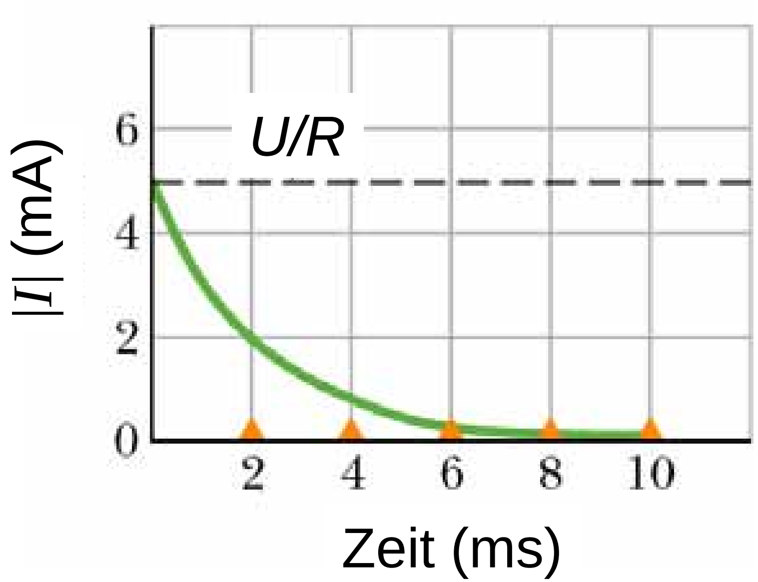
Halliday-Resnick-Walker (2003) Abb. 26.14

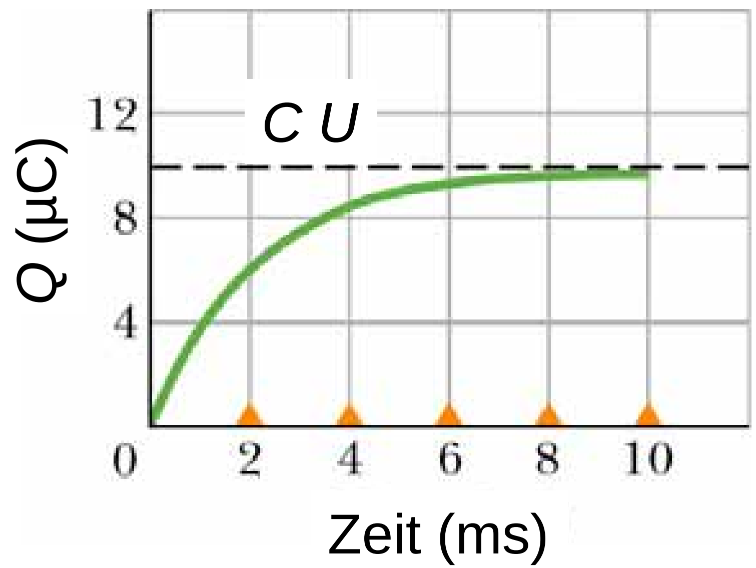
zu 2.4: Laden eines Kondensators



Halliday-Resnick-Walker (2003) Abb. 26.13

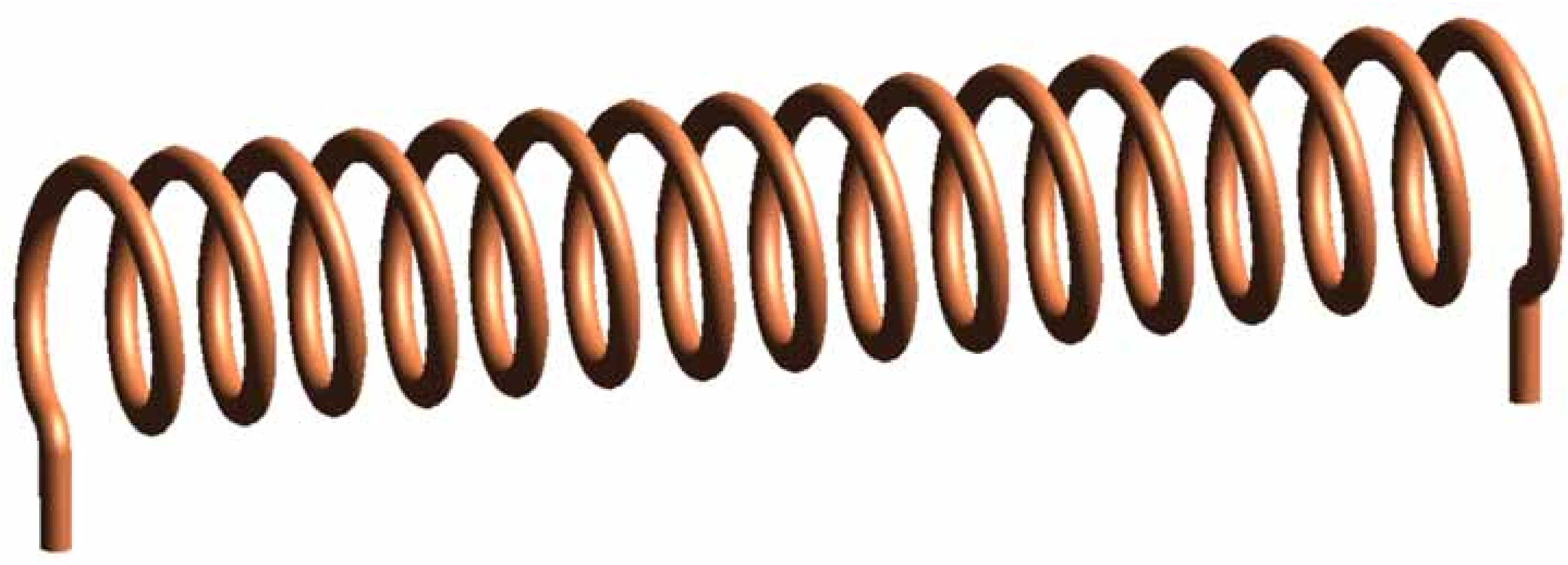
zu 2.4: Laden eines Kondensators





Halliday-Resnick-Walker (2003) Abb. 26.14

zu 2.5: Zylinderspule

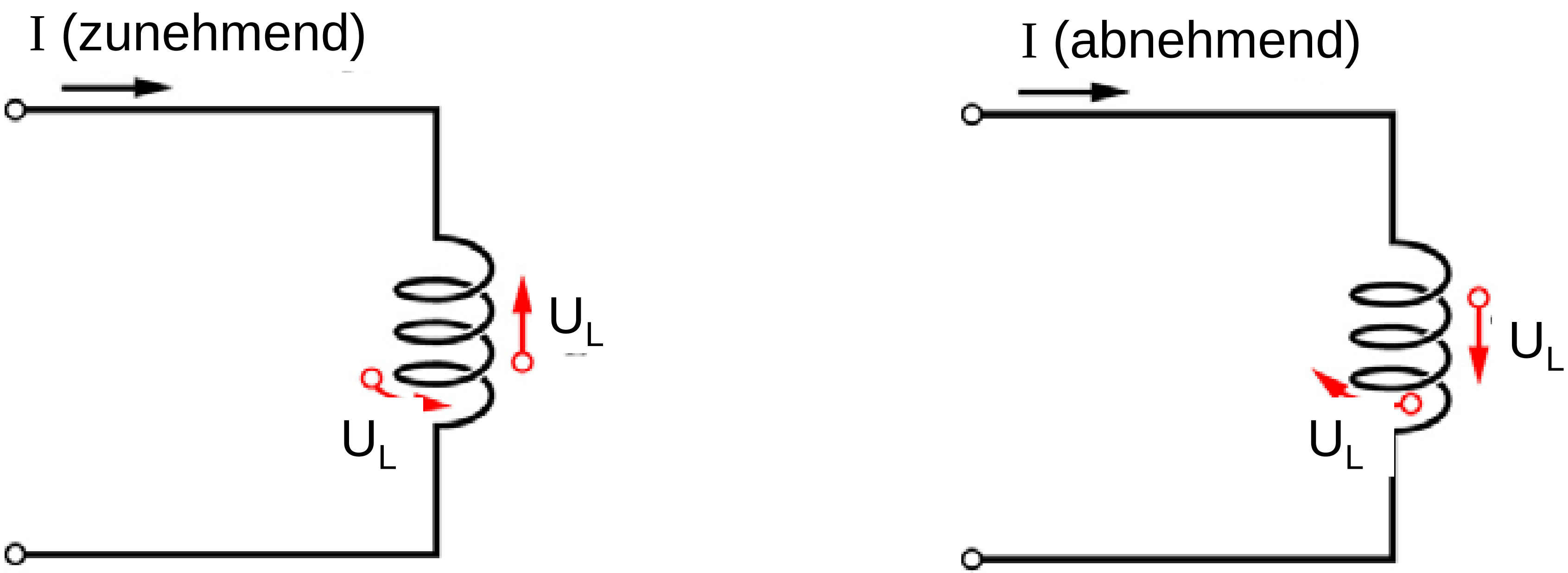


Wikipedia.org

zu 2.5: Induktivitäten

rn-wissen.de

zu 2.5: Induktivität einer Spule

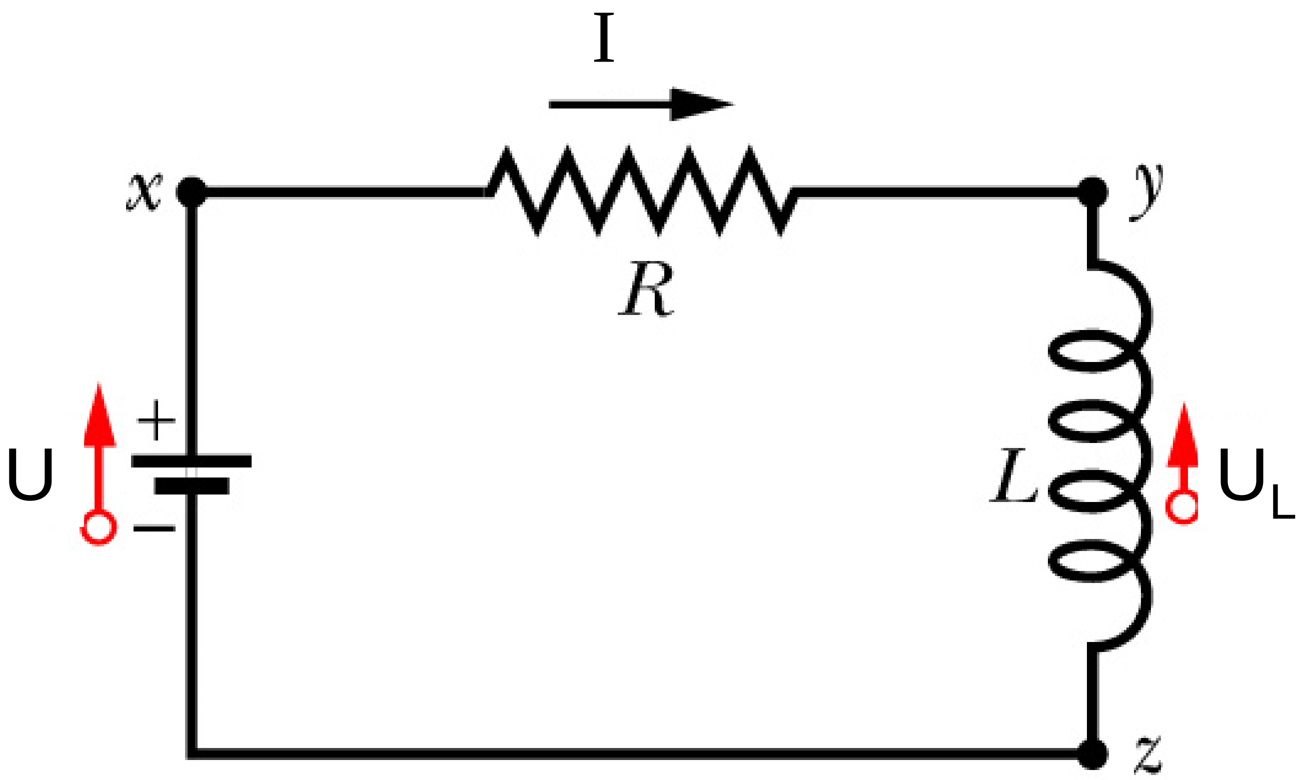


$$U = -L\frac{dI}{dt} = -U_L$$

Halliday-Resnick-Walker (2003) Abb. 31.16

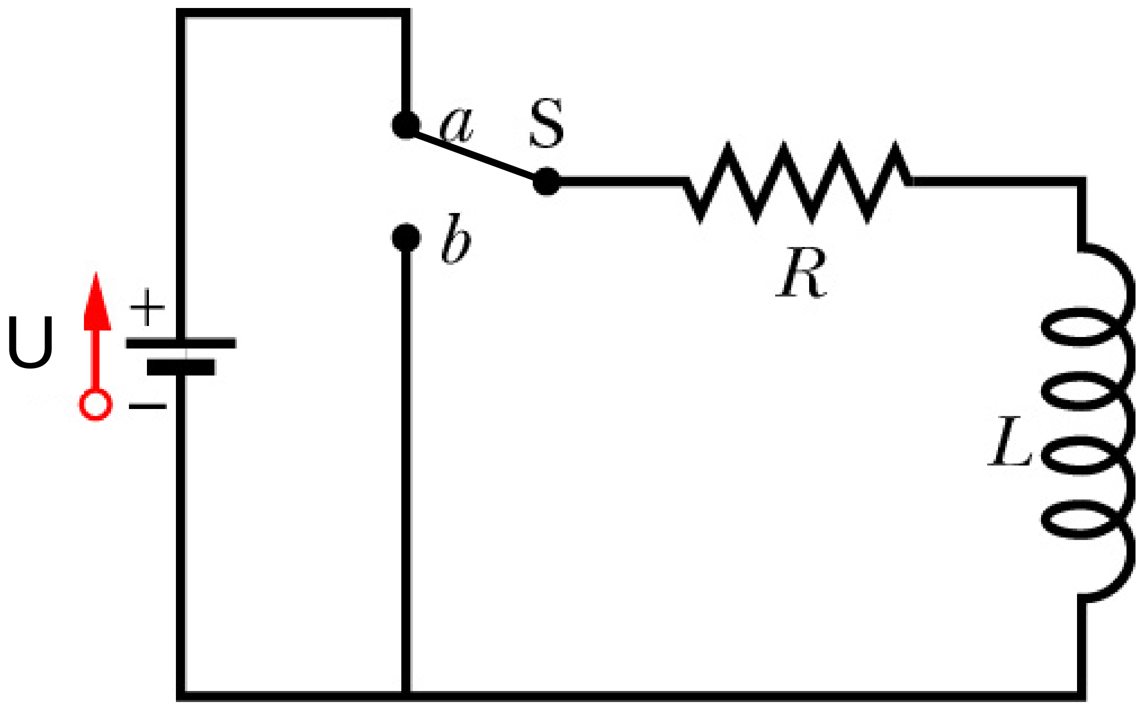
zu 2.5: RL-Kreis

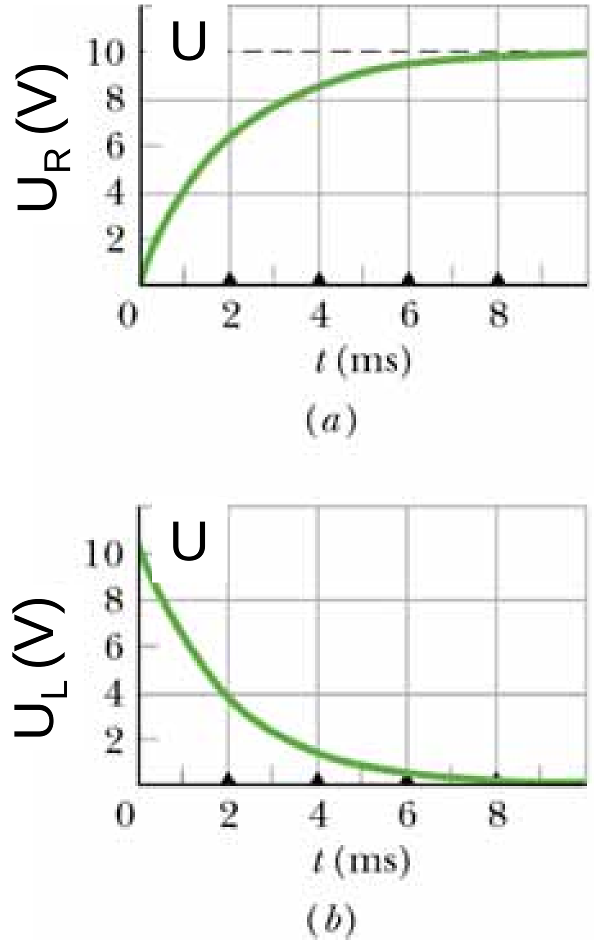
$$U = RI + L\frac{dI}{dt}$$
$$= U_R + U_L$$



Halliday-Resnick-Walker (2003) Abb. 31.18

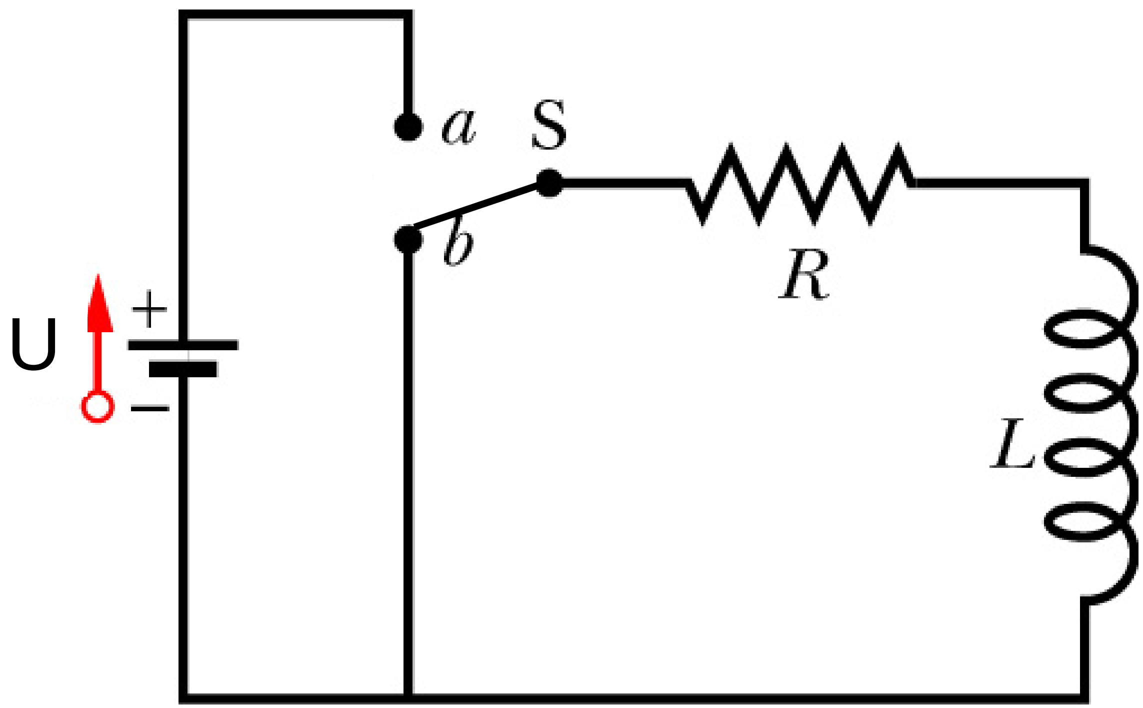
zu 2.5: RL-Kreis - Einschaltvorgang





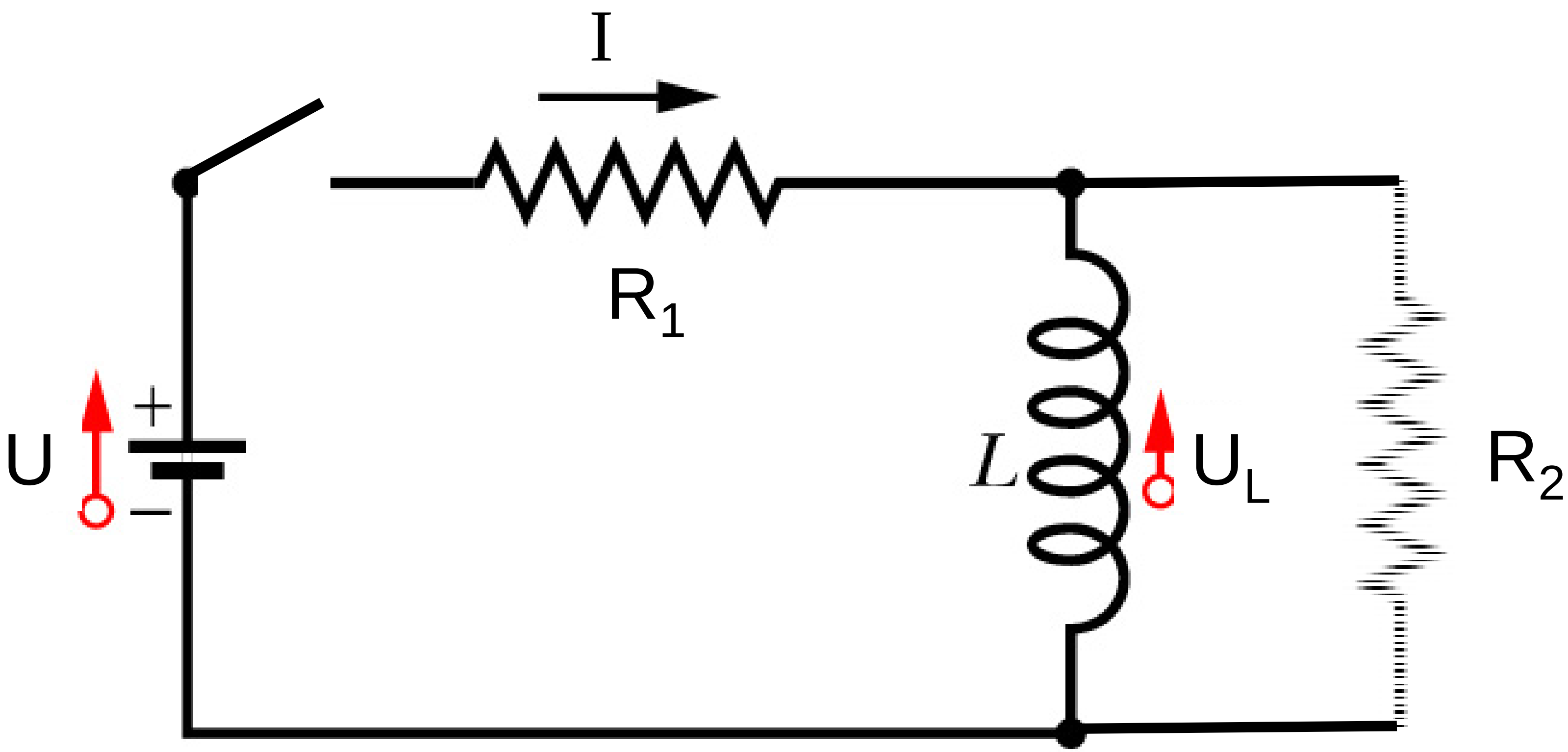
Halliday-Resnick-Walker (2003) Abb. 31.17 und 31.19

zu 2.5: RL-Kreis - Ausschaltvorgang



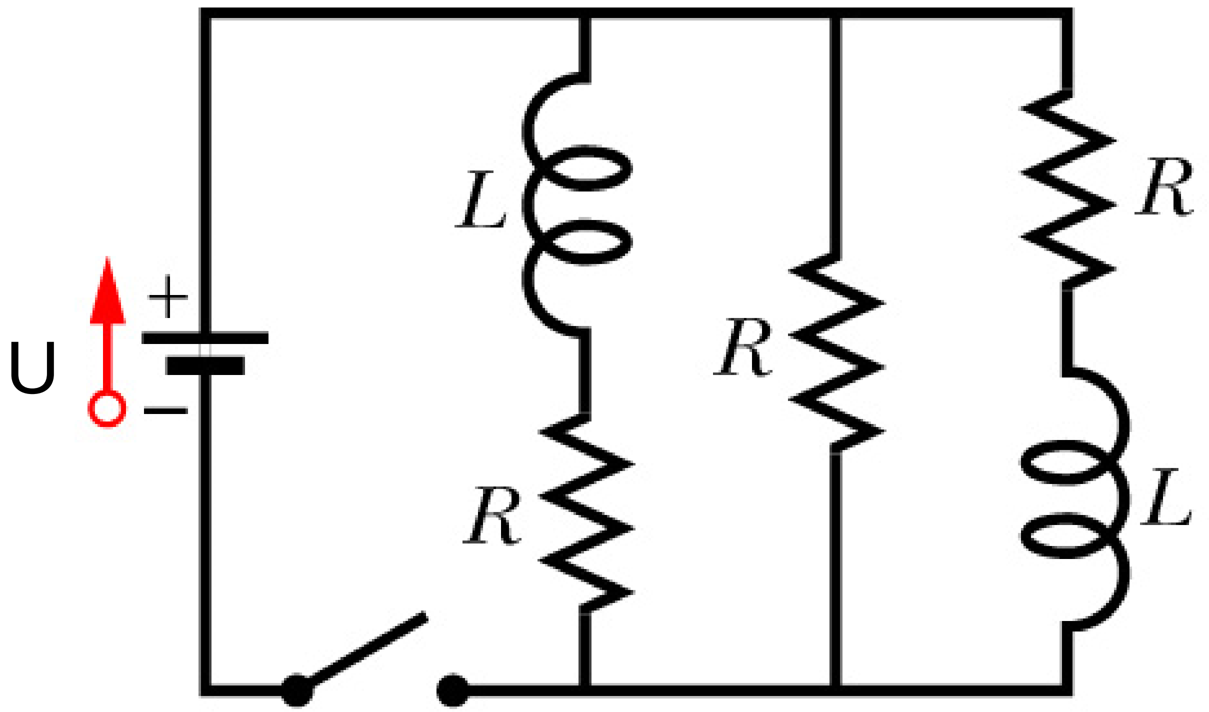
Halliday-Resnick-Walker (2003) Abb. 31.17 und 31.19

zu 2.5: Selbstinduktion beim Abschalten einer Spule



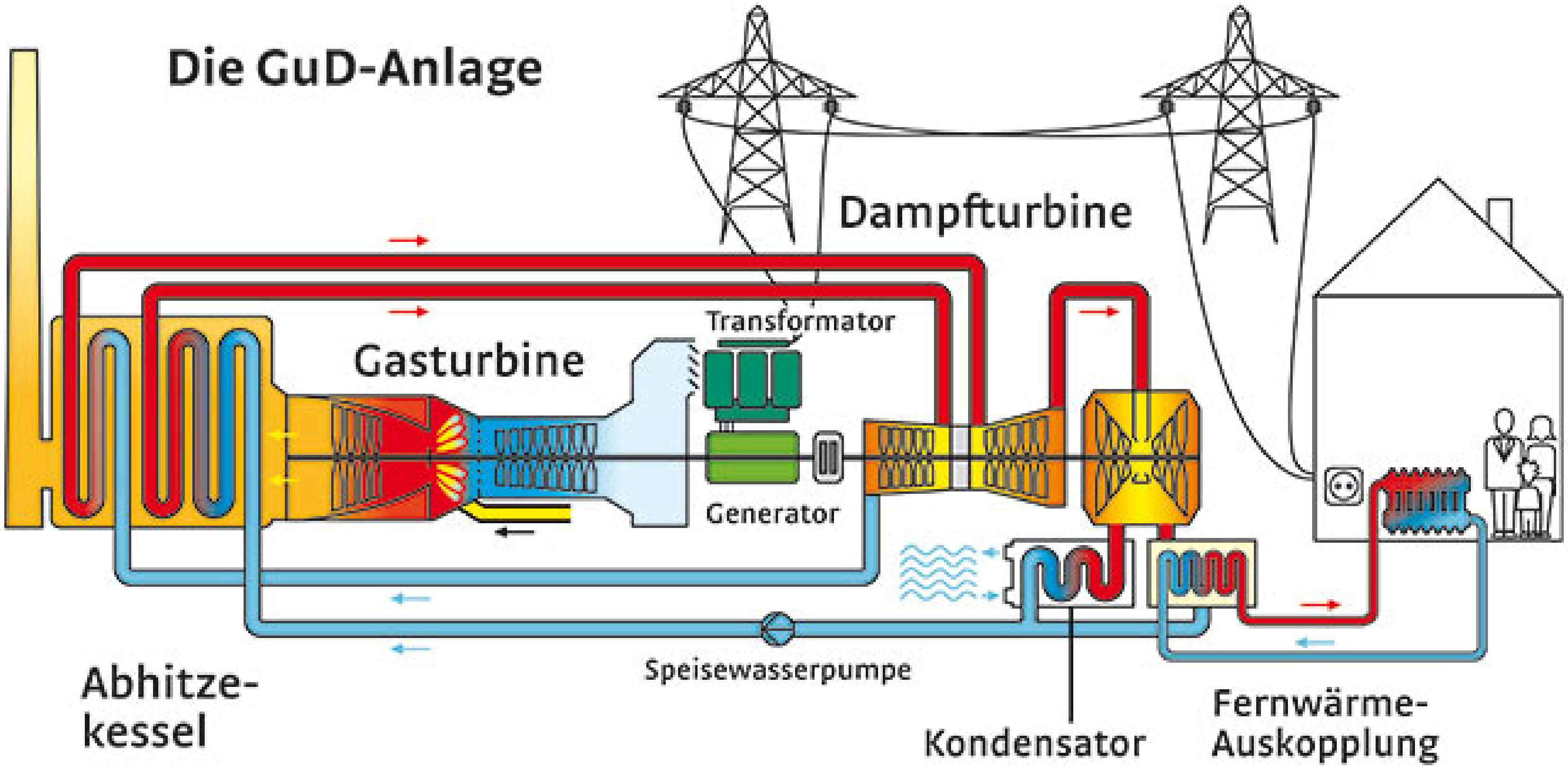
Halliday-Resnick-Walker (2003) Abb. 31.18

zu 2.5: Beispiel für einen RL-Kreis



Halliday-Resnick-Walker (2003) Abb. 31.20

zu 2.6: Generator



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zu 2.6: Generator

Dampfturbine

Generator

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zu 2.6: Wechselspannungsquelle

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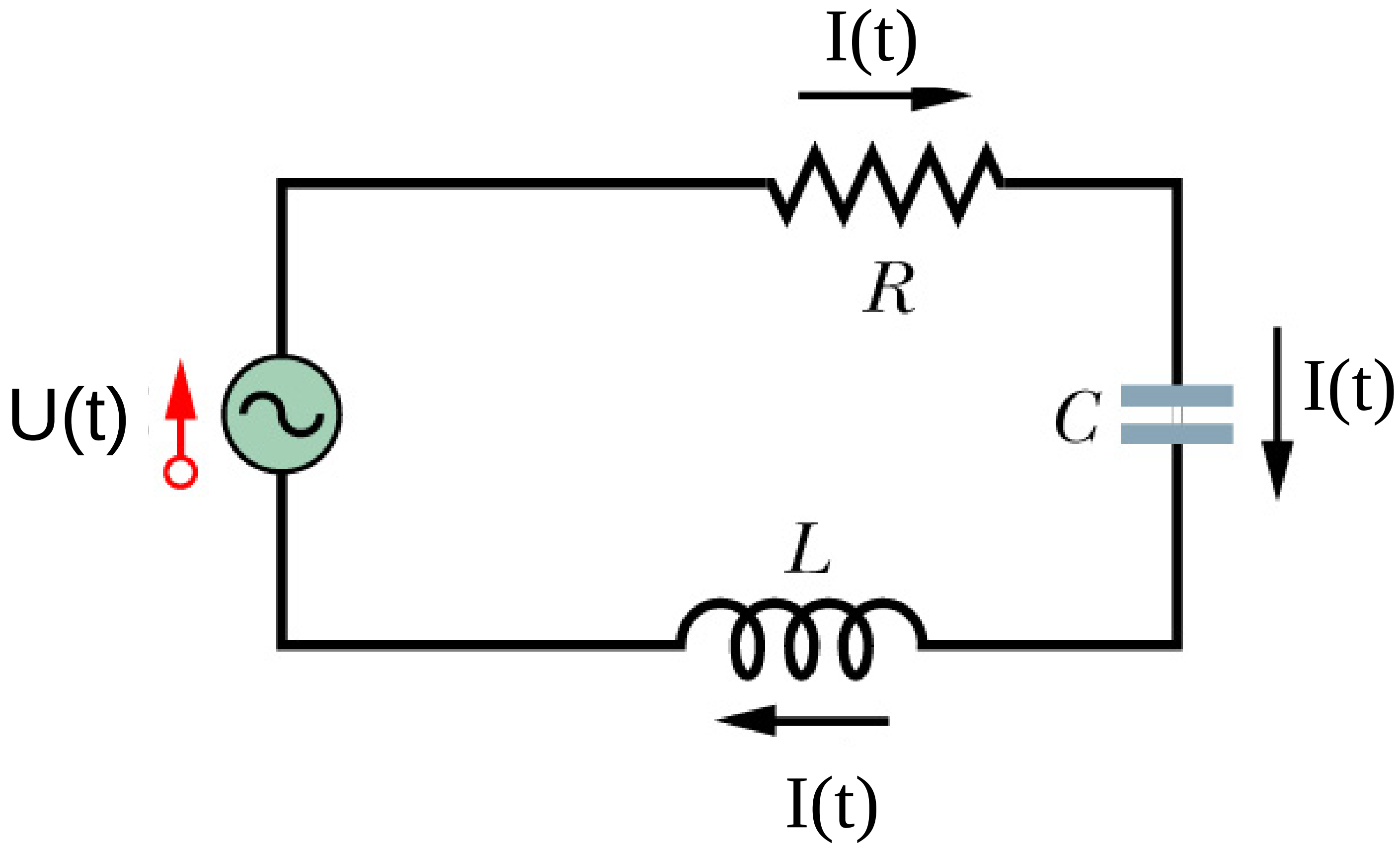
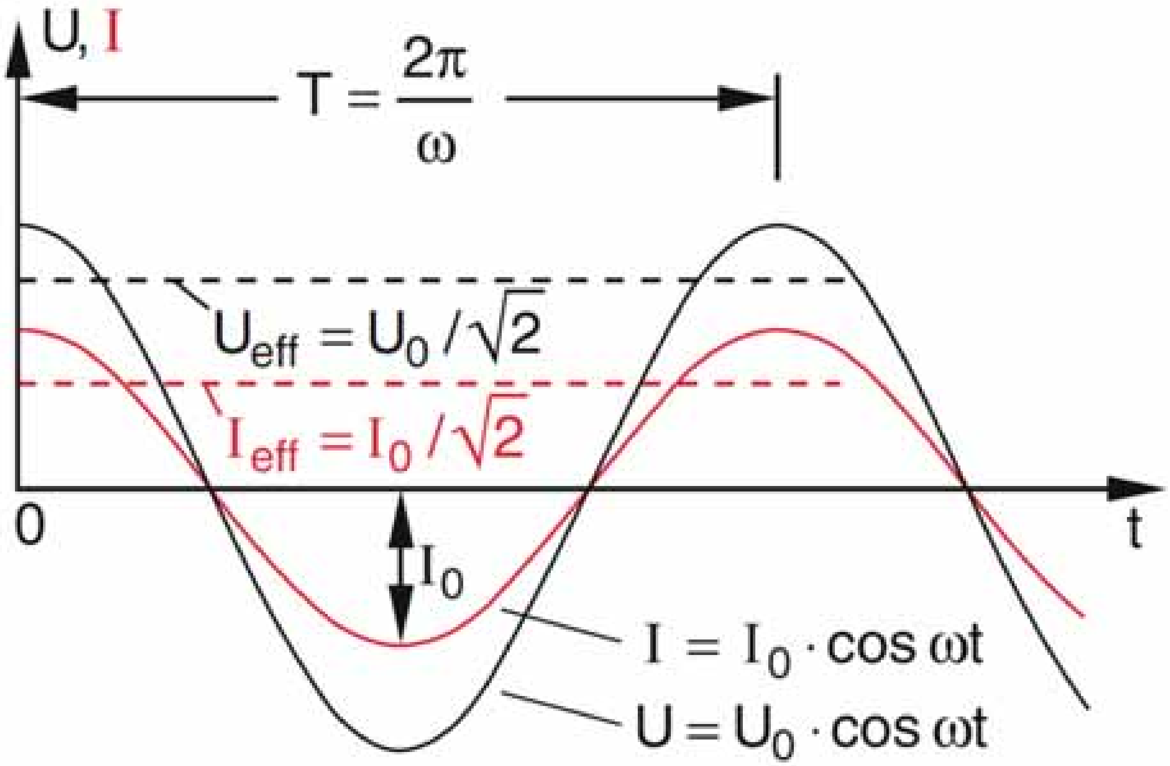
zu 2.6: Wechselstromkreise



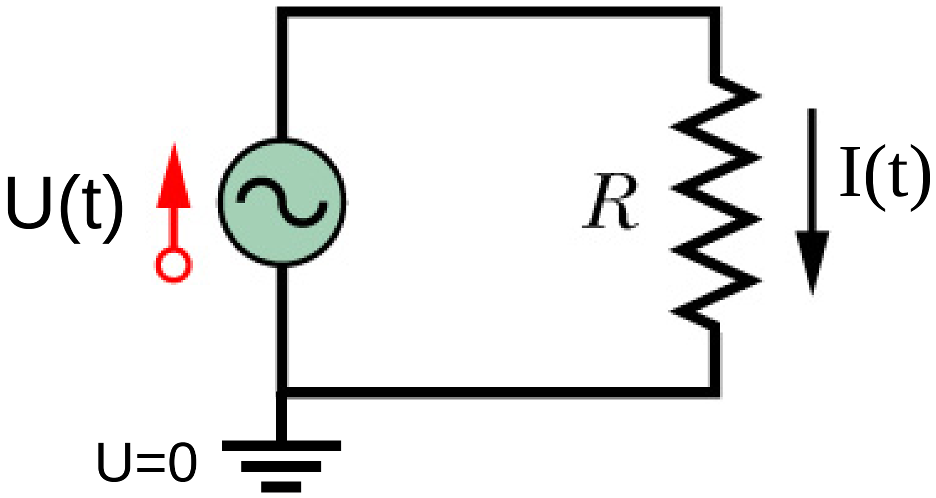
Halliday-Resnick-Walker (2003) Abb. 33.7

zu 2.6: Wechselstromkreise

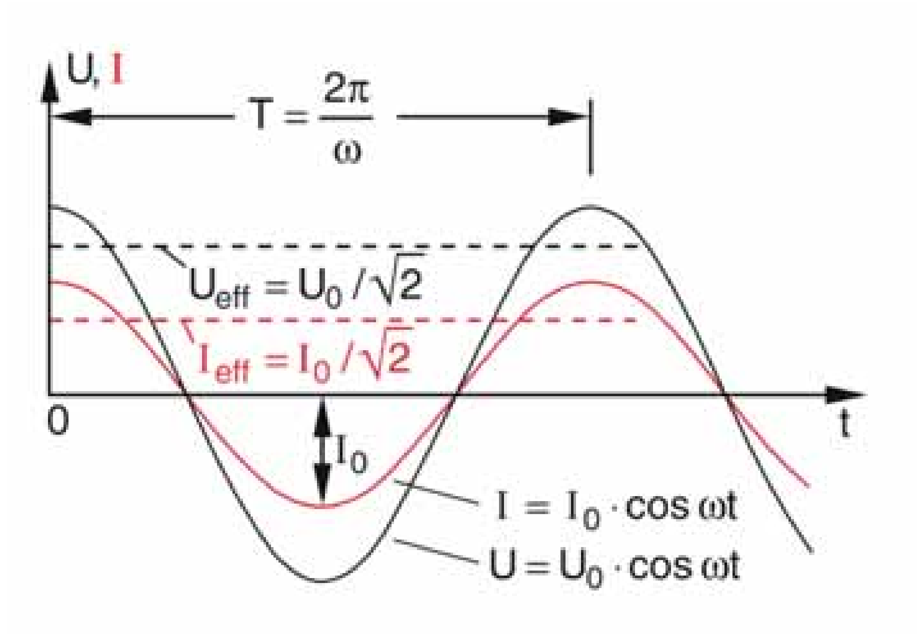


Demtröder – Experimentalphysik 2 (2013), Abb. 5.11.

zu 2.6: R im Wechselstromkreis

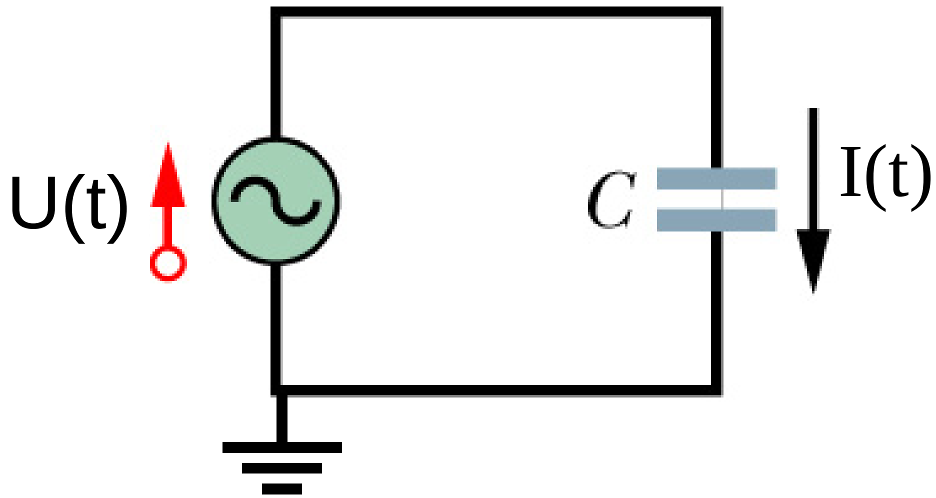


Halliday-Resnick-Walker (2003) Abb. 33.8

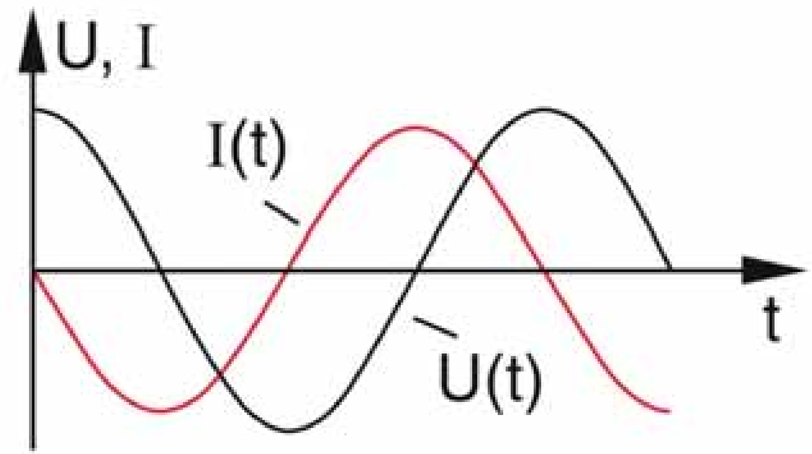


Demtröder – Experimentalphysik 2 (2013), Abb. 5.11.

zu 2.6: C im Wechselstromkreis

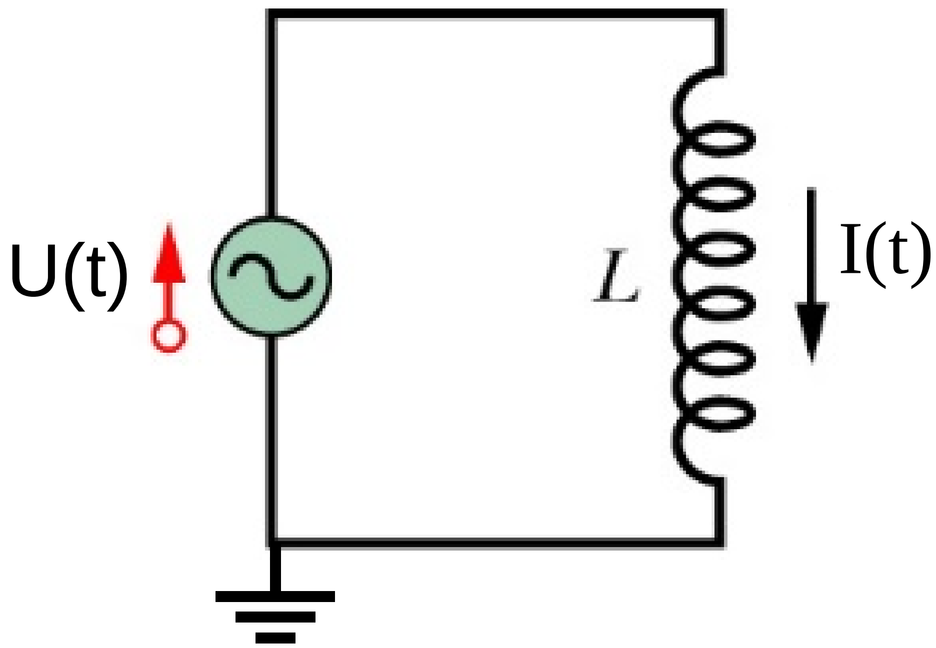


Halliday-Resnick-Walker (2003) Abb. 33.9

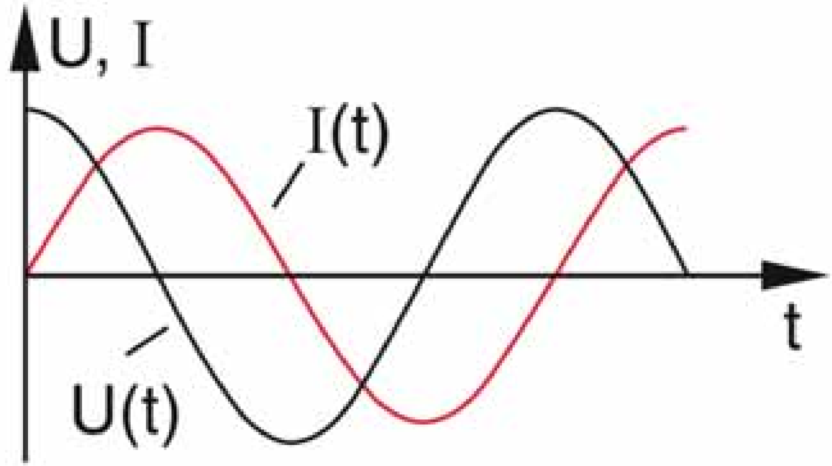


Demtröder – Experimentalphysik 2 (2013), Abb. 5.25.

zu 2.6: L im Wechselstromkreis



Halliday-Resnick-Walker (2003) Abb. 33.9



Demtröder – Experimentalphysik 2 (2013), Abb. 5.23.

zu 2.6: Impedanz

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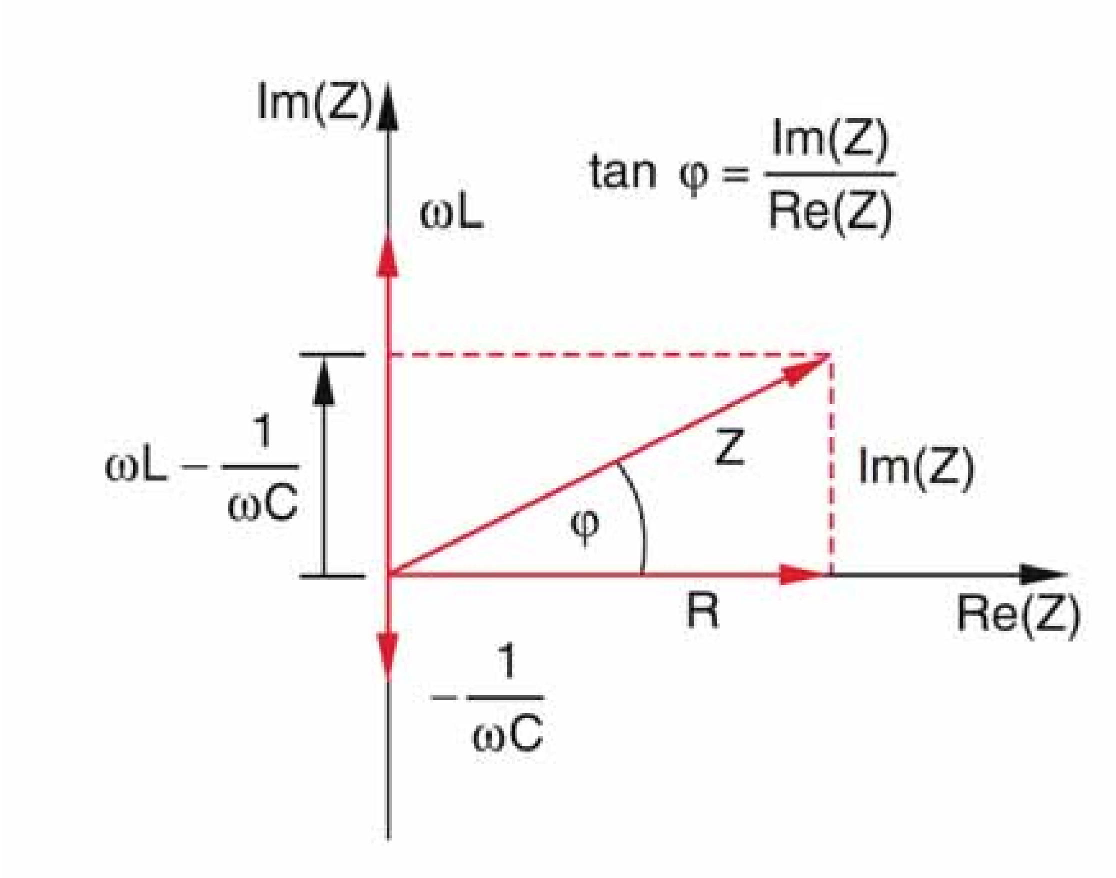
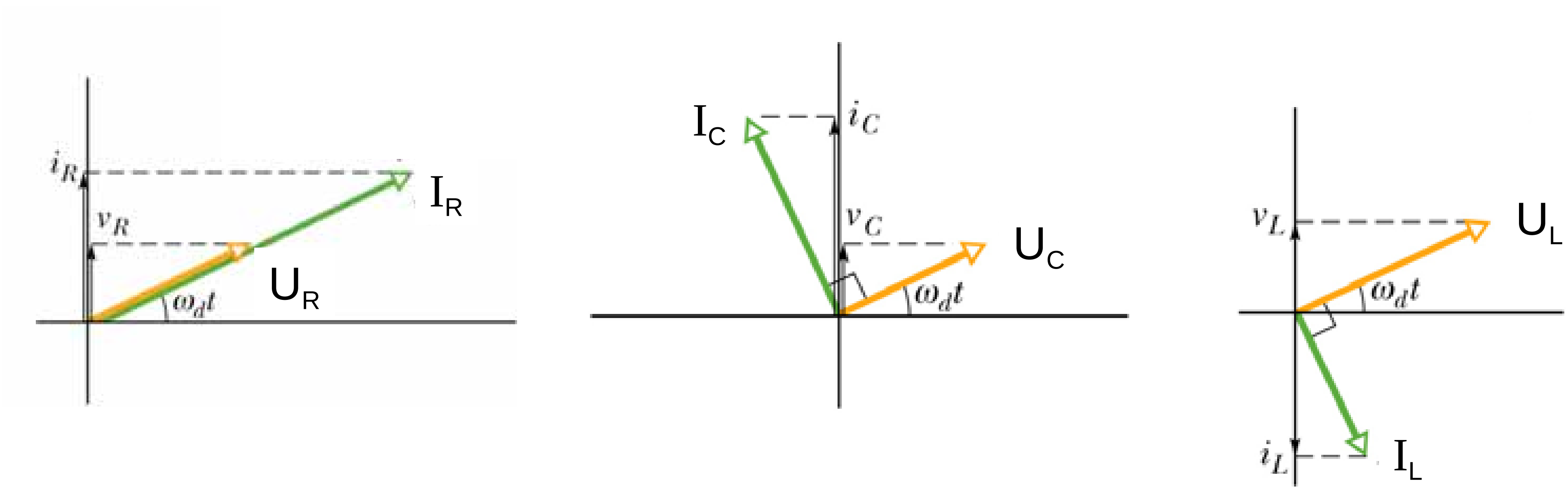


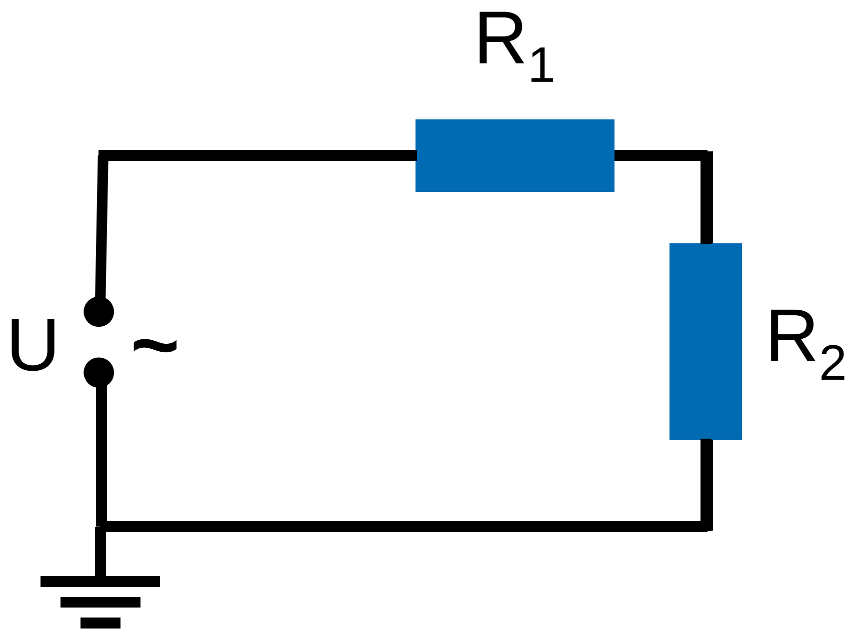
Demtröder – Experimentalphysik 2 (2013), Abb. 5.27.

zu 2.6: Zeigerdiagramm



Halliday-Resnick-Walker (2003) Abb. 33,7, 33.8, 33.9

zu 2.6: Spannungsteiler



zu 2.6: Hochpass

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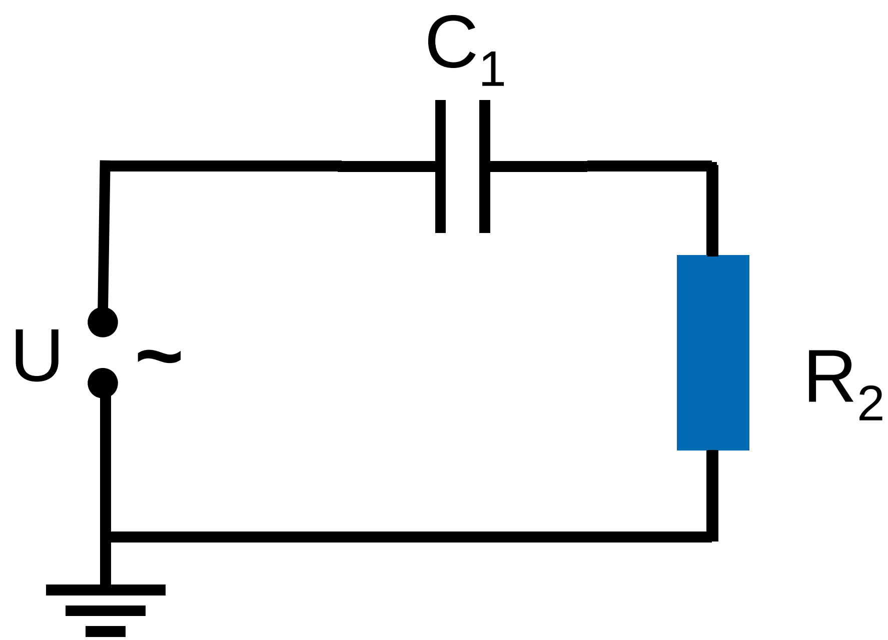
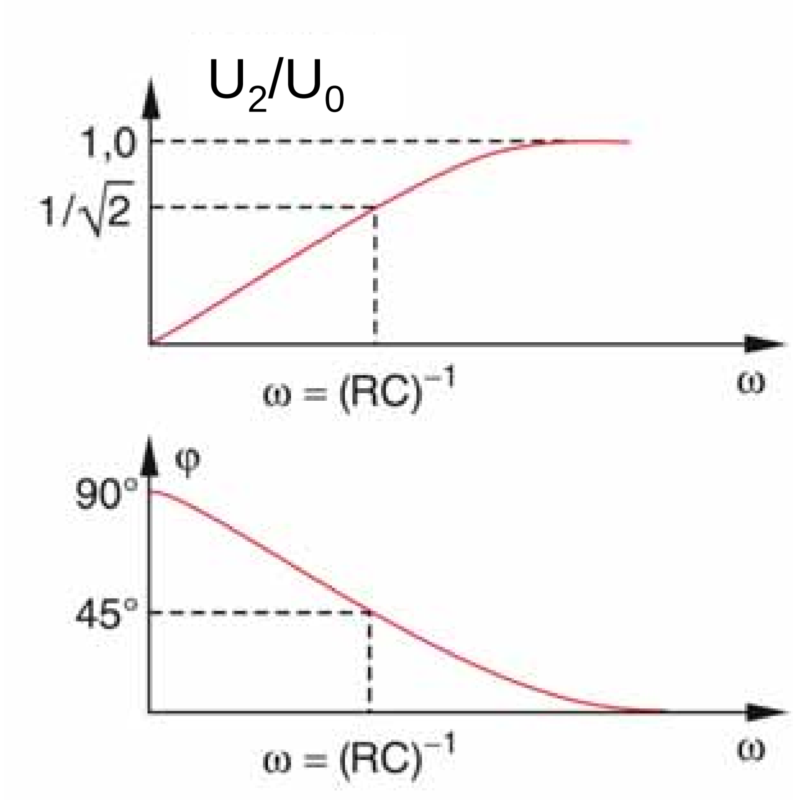
CR-Hochpassfilter

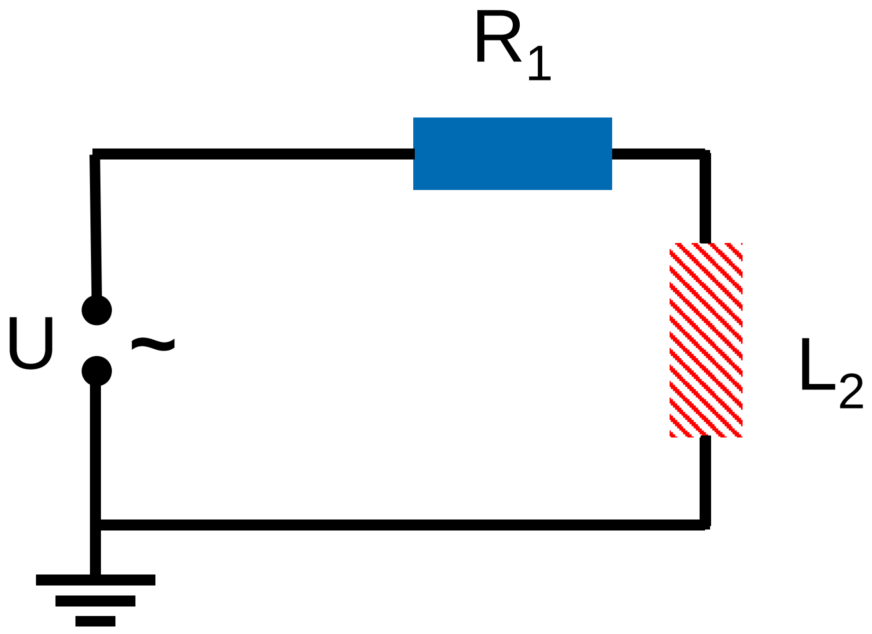




Demtröder – Experimentalphysik 2 (2013), Abb. 5.29.

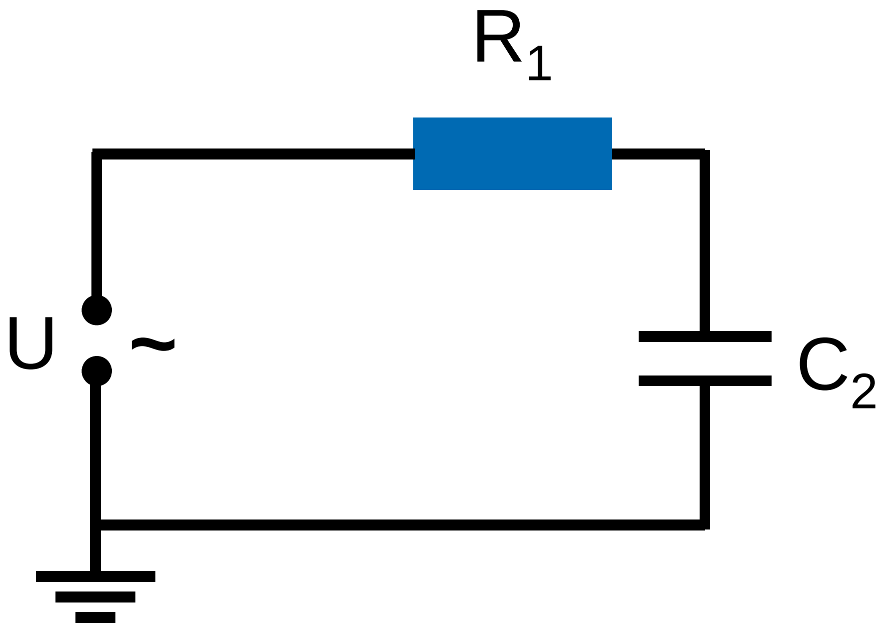
zu 2.6: Hochpassfilter

RL-Hochpassfilter



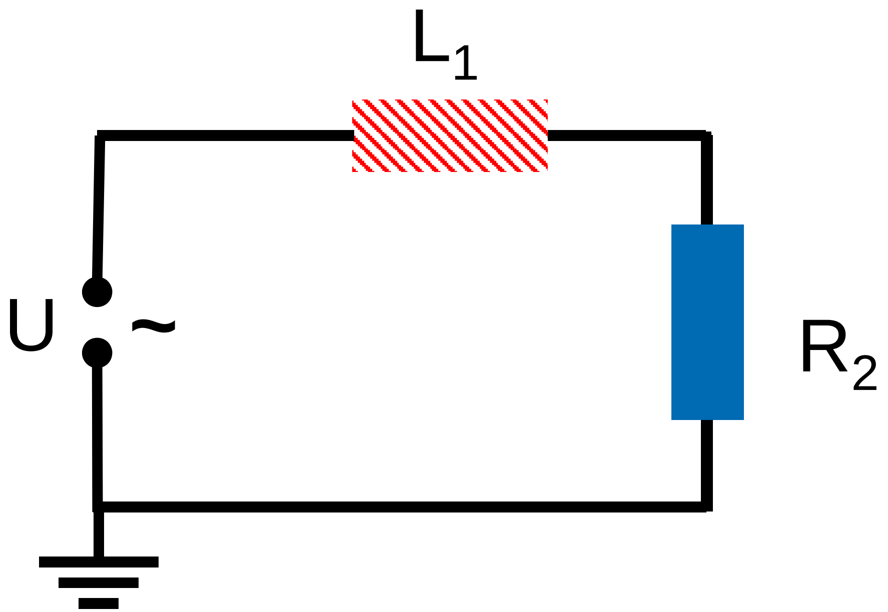
zu 2.6: Tiefpass

RC-Tiefpassfilter



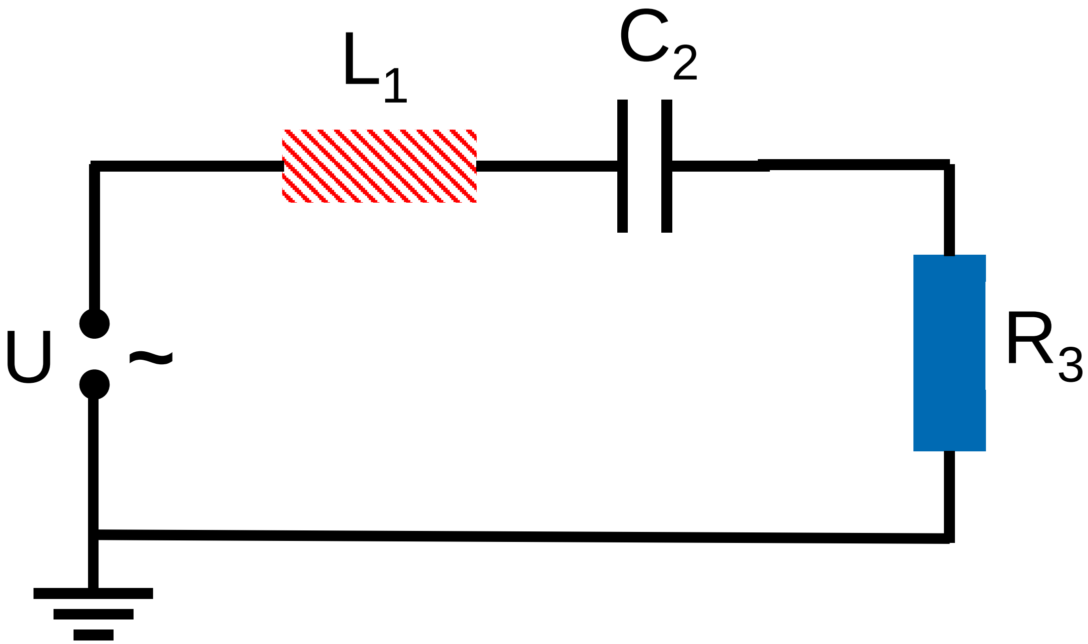
zu 2.6: Tiefpass

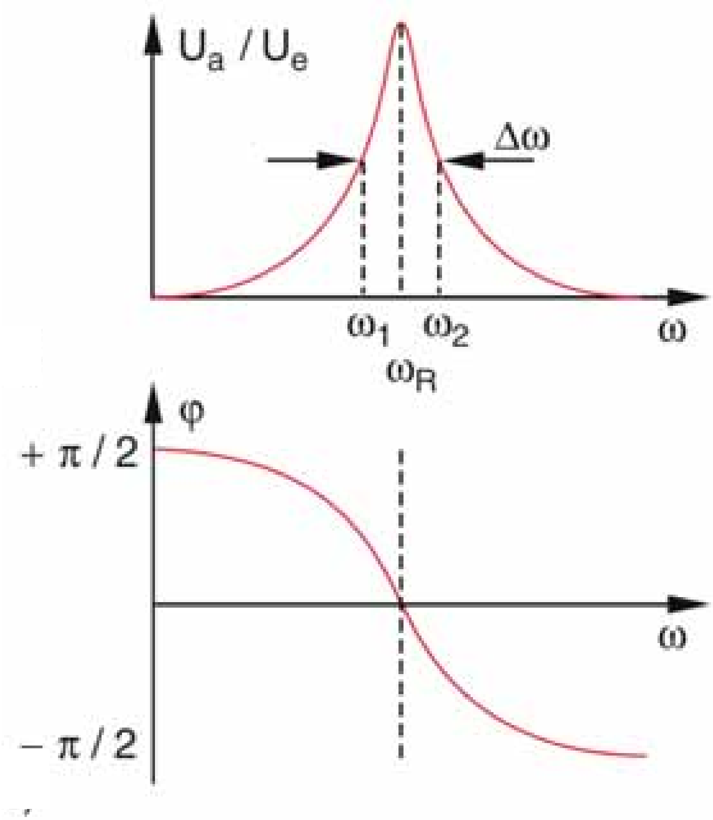
LR-Tiefpassfilter



zu 2.6: Frequenzfilter

LCR-Durchlassfilter

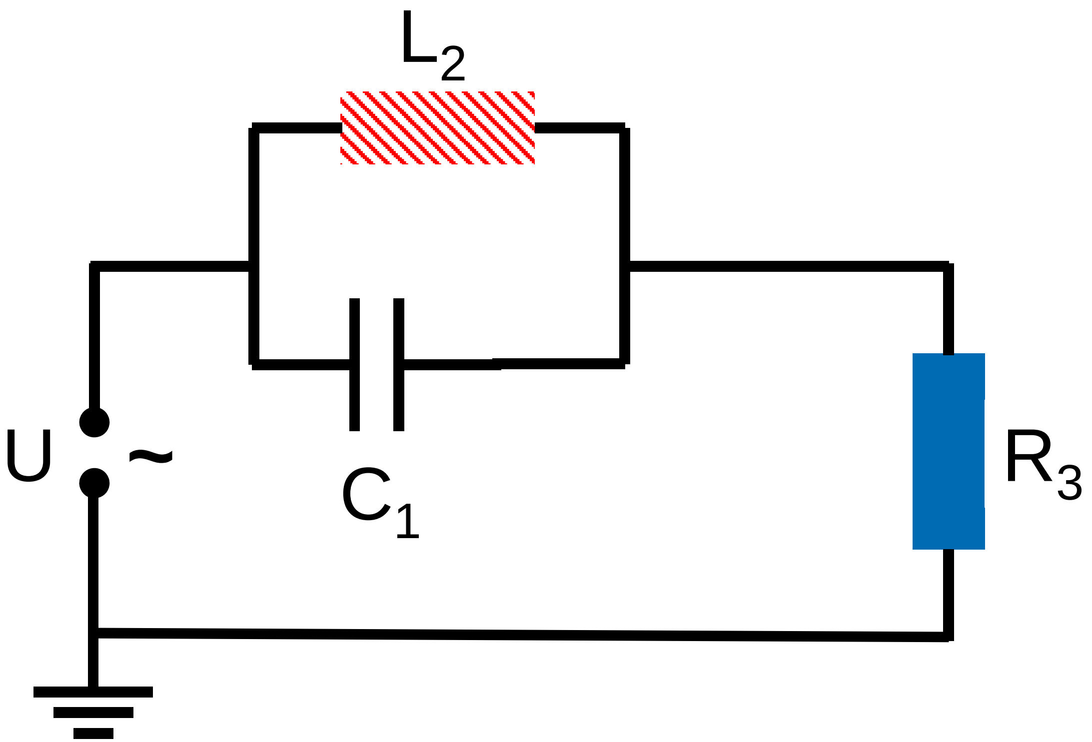


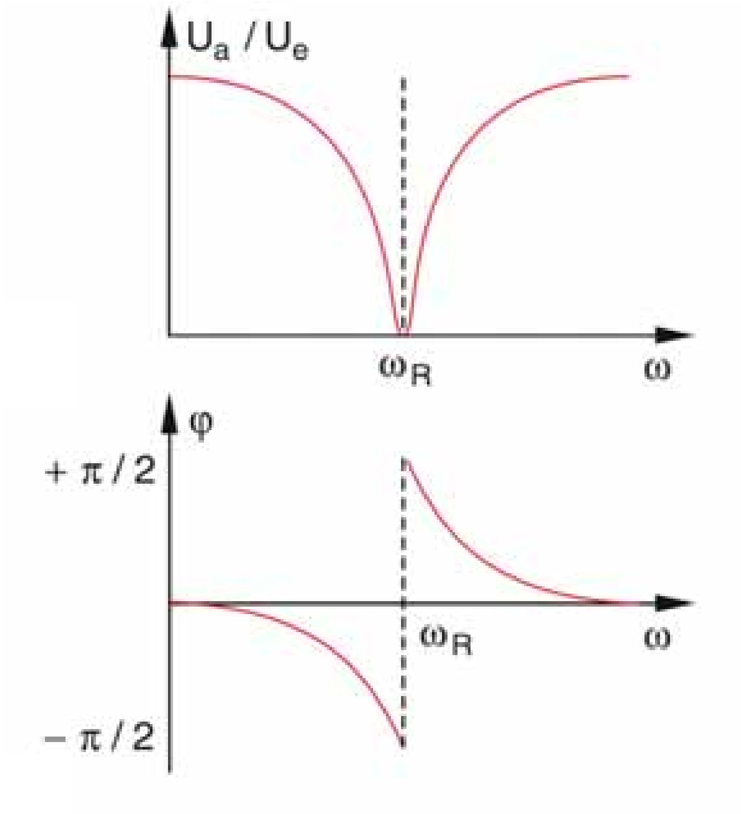


Demtröder – Experimentalphysik 2 (2013), Abb. 5.31.

zu 2.6: Frequenzfilter

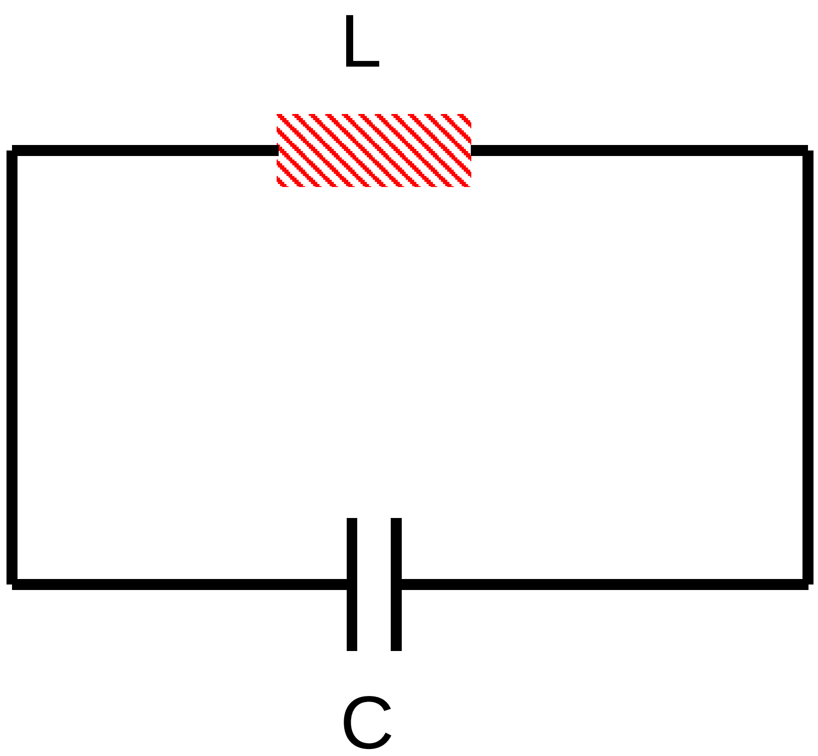
LCR-Sperrfilter



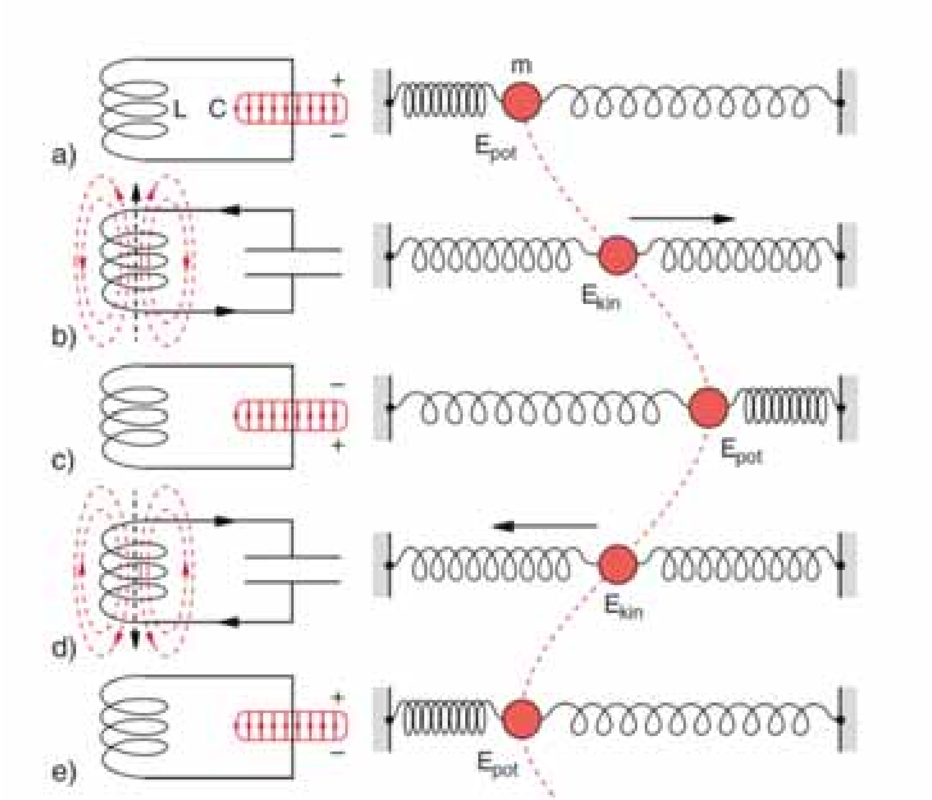


Demtröder – Experimentalphysik 2 (2013), Abb. 5.32.

zu 2.7: LC-Schwingungen

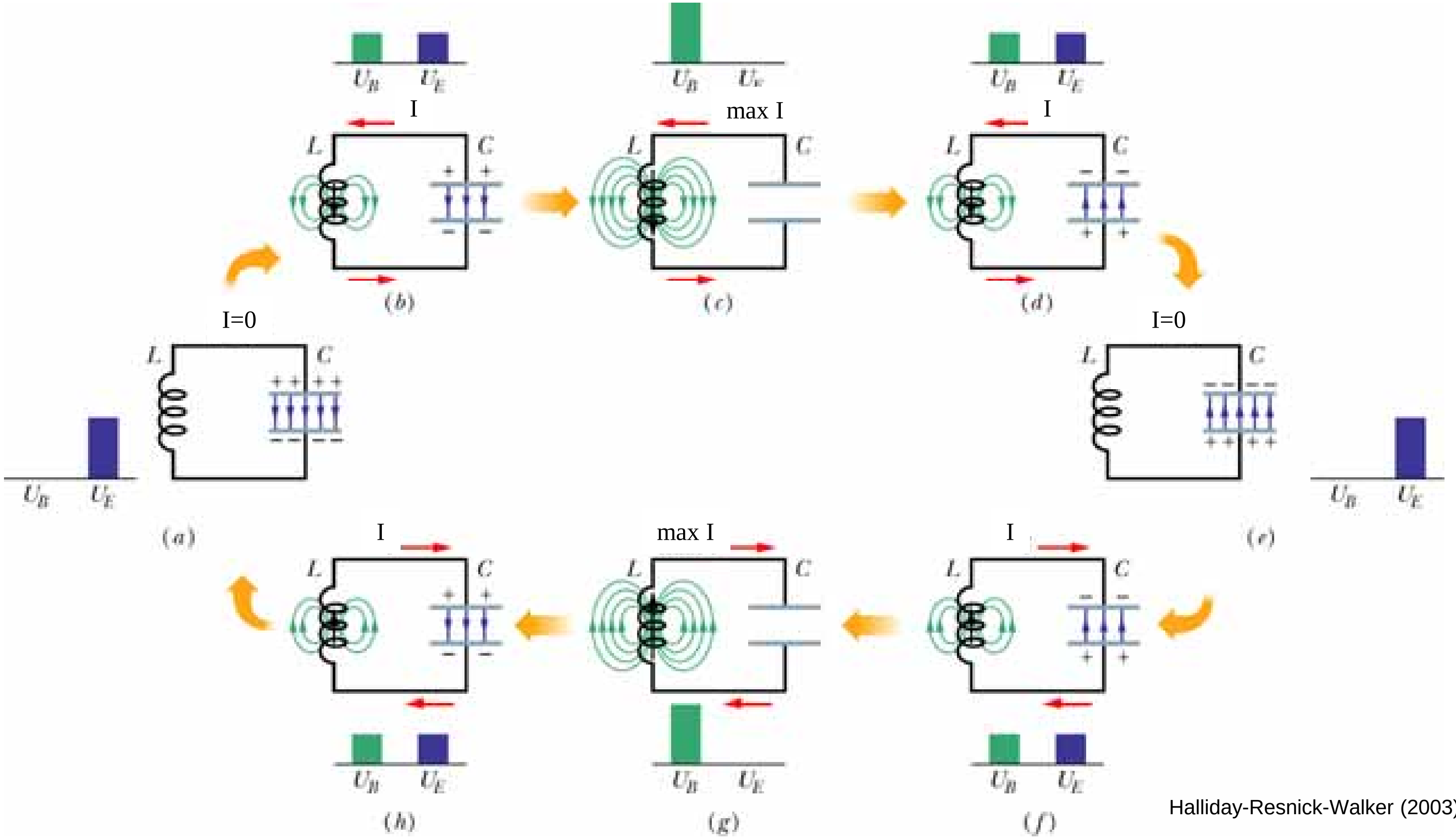


zu 2.7: LC-Schwingungen



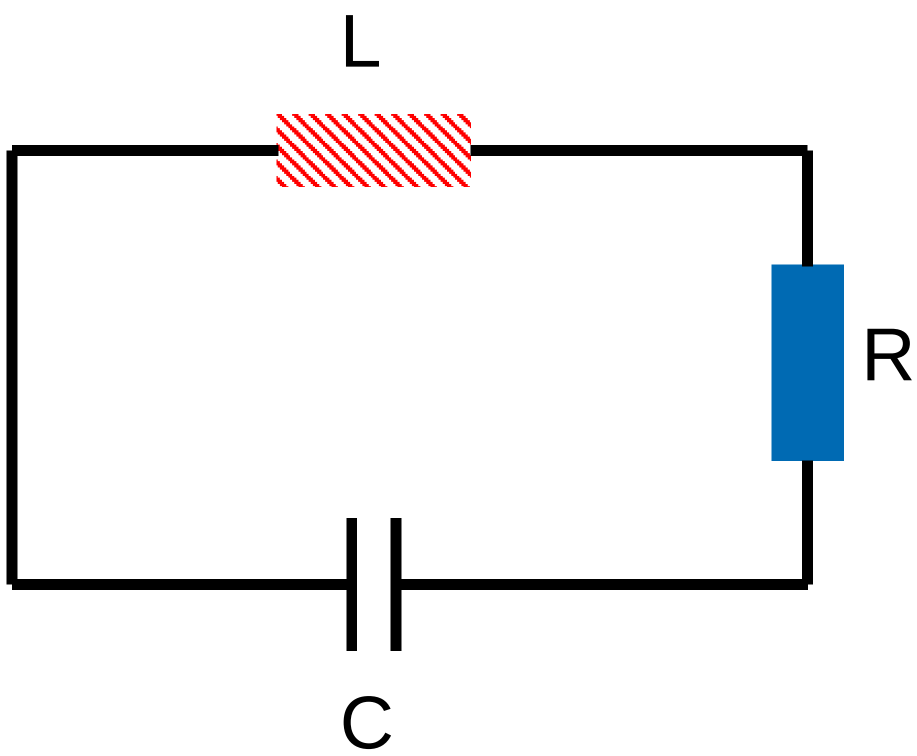
Demtröder – Experimentalphysik 2 (2013), Abb. 6.31.

zu 2.7: LC-Kreis

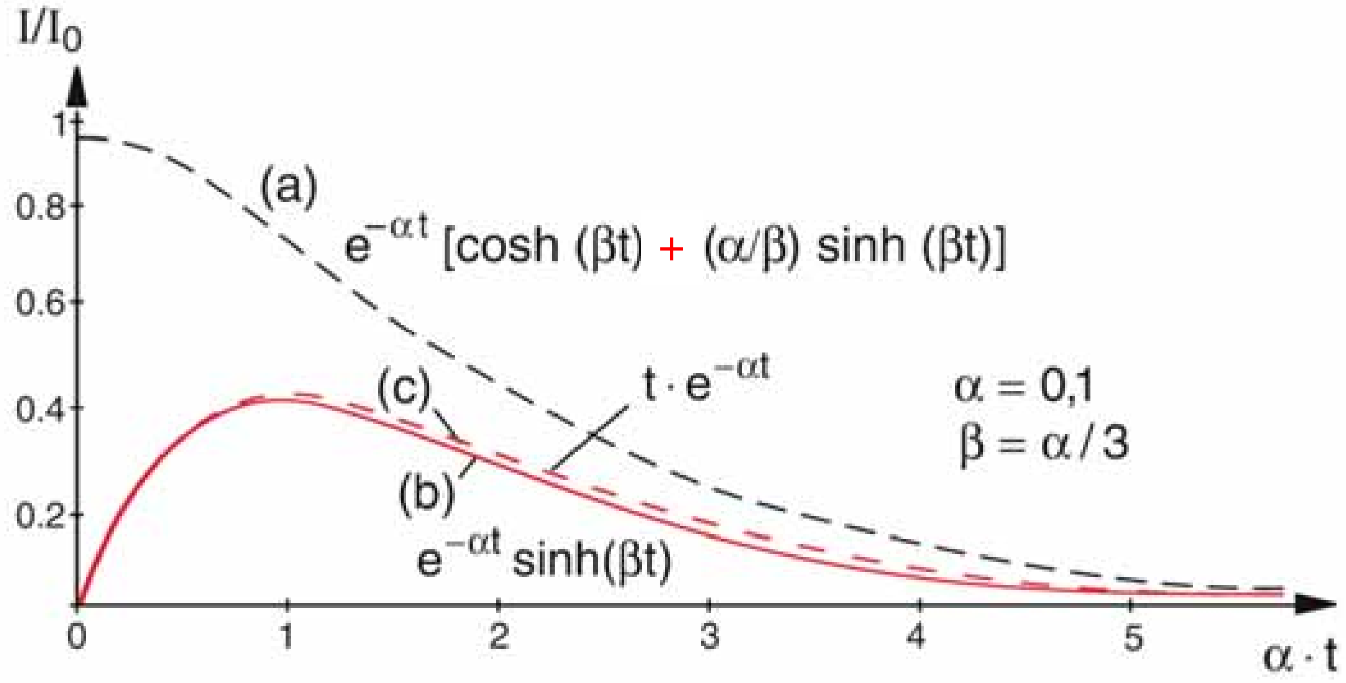


Halliday-Resnick-Walker (2003) Abb. 33.1

zu 2.7: RLC-Kreis



zu 2.7: RLC-Kreis



Demtröder – Experimentalphysik 2 (2013), Abb. 6.3.

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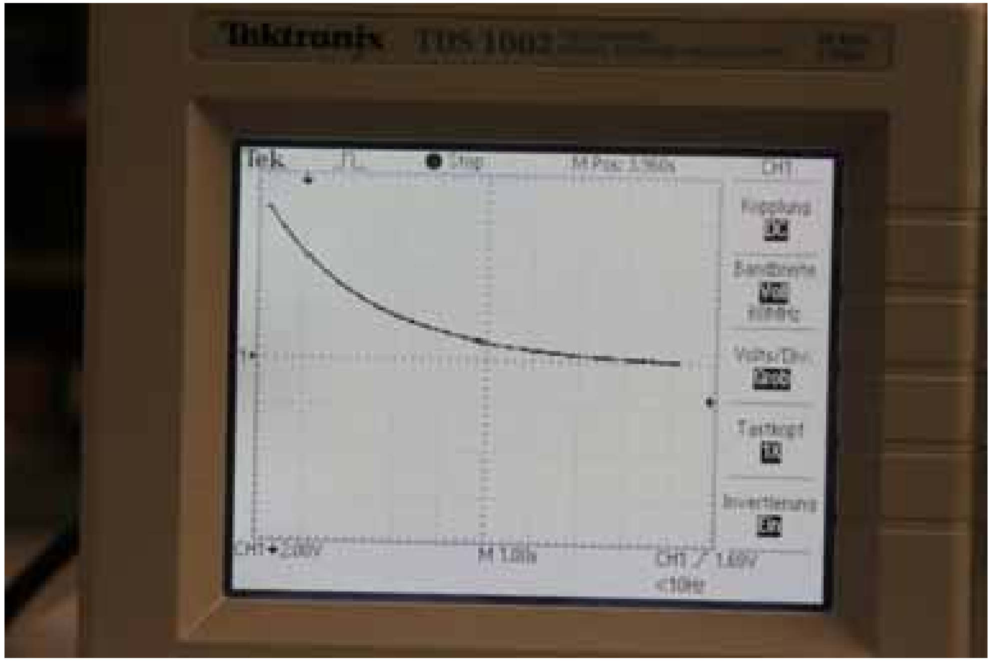
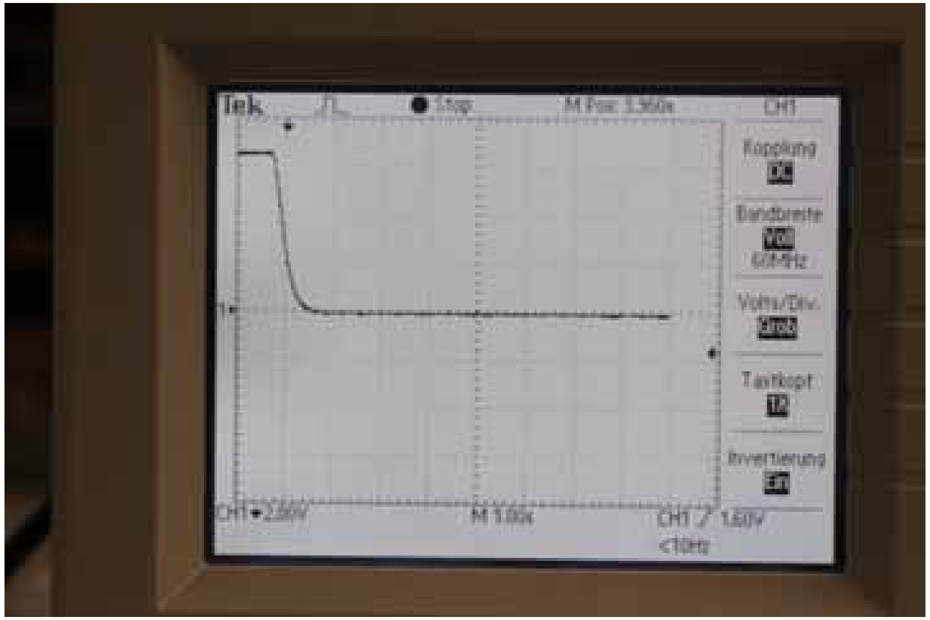
Kriechfall



$$U_C(t) = U_0\, \mathrm{e}^{-\delta\, t} \left[\cosh\left(\beta\; t\right) + \frac{\delta}{\beta} \sinh\left(\beta\; t\right)\right]$$

zu 2.7: RLC-Kreis

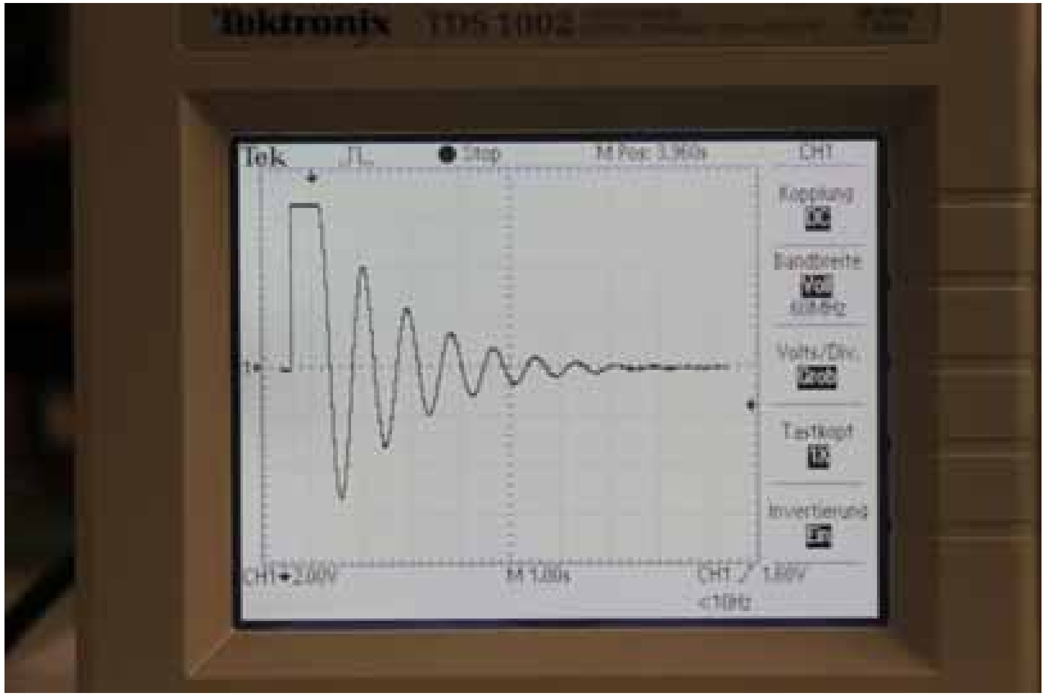
aperiodischer Grenzfall



$$U_C(t) = U_0\, \mathrm{e}^{-\delta\, t}\, [1 + \delta\, t]$$

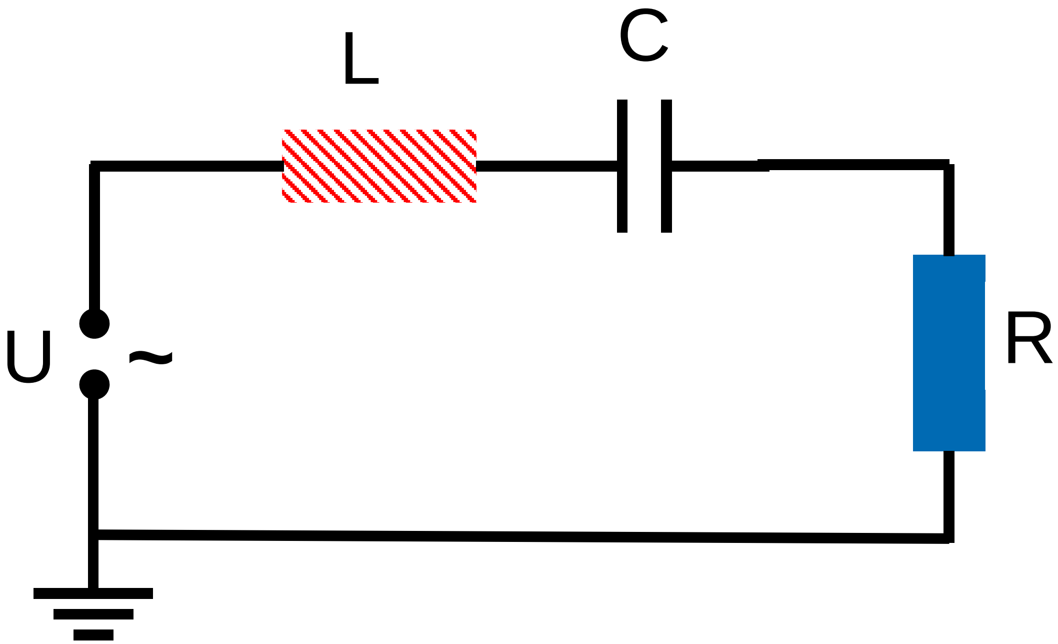
zu 2.7: RLC-Kreis

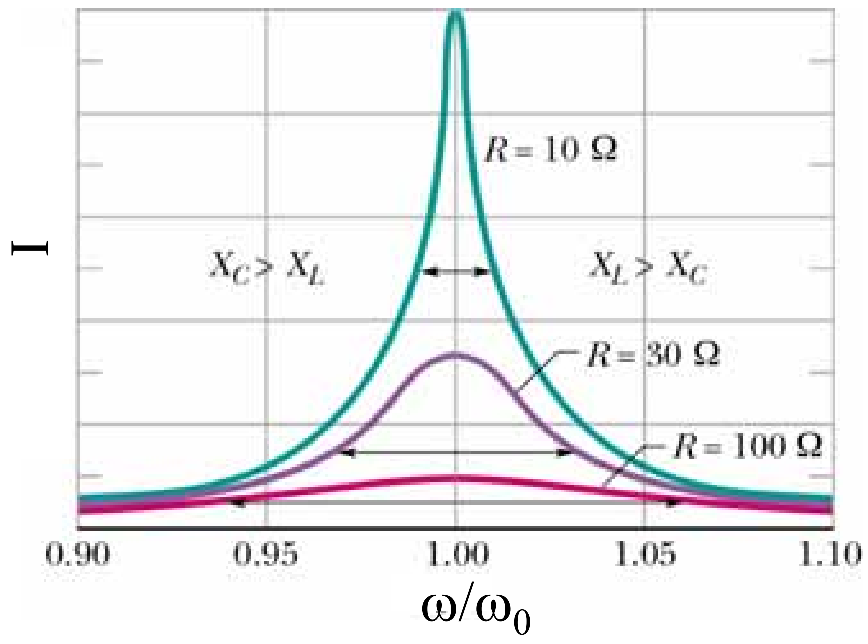
Schwingfall



$$U_C(t) \quad = \quad U_0\, \mathrm{e}^{-\delta\, t}\, \cos\left(\omega\, t\right)$$

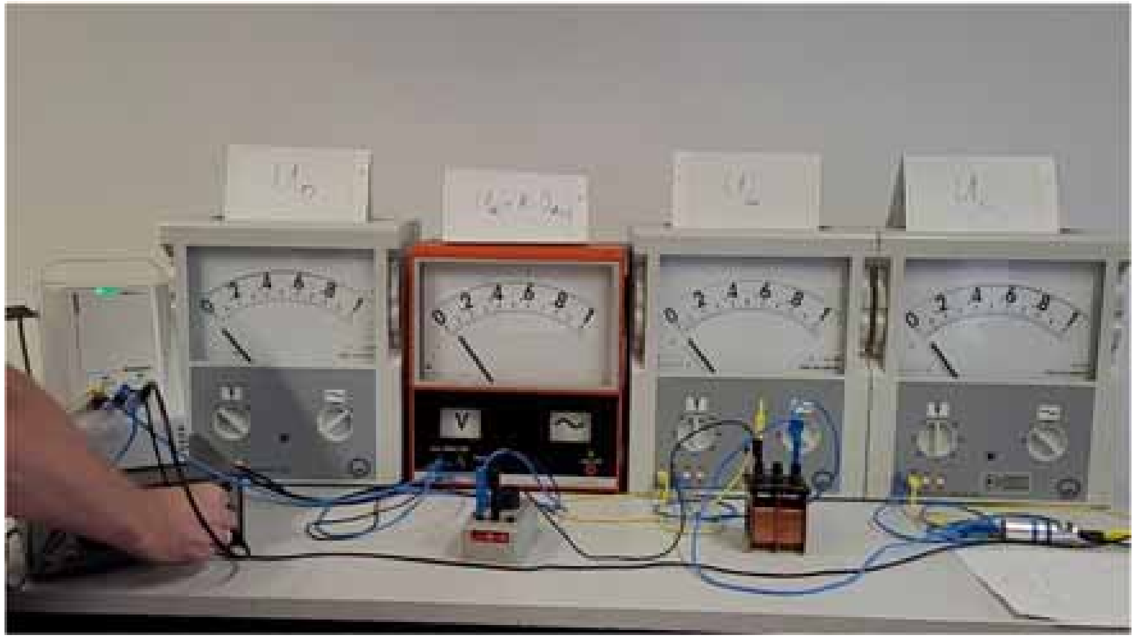
zu 2.7: Erzwungene Schwingung

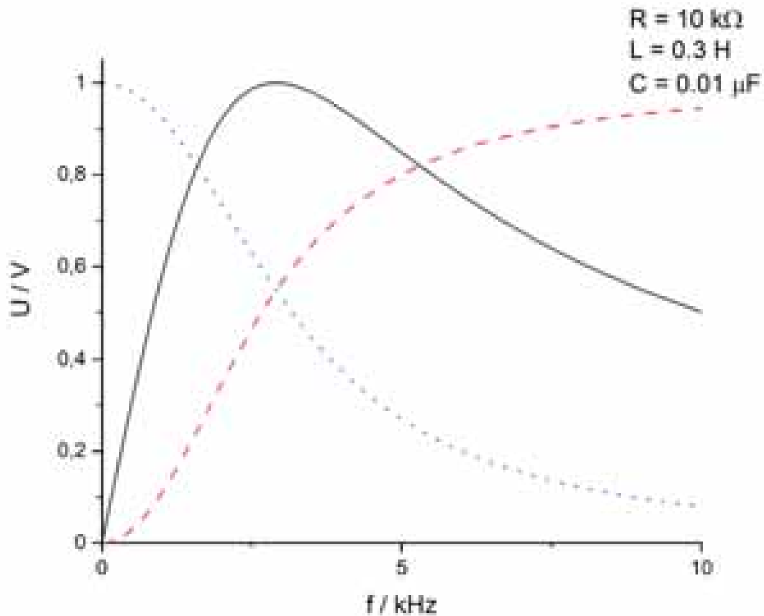




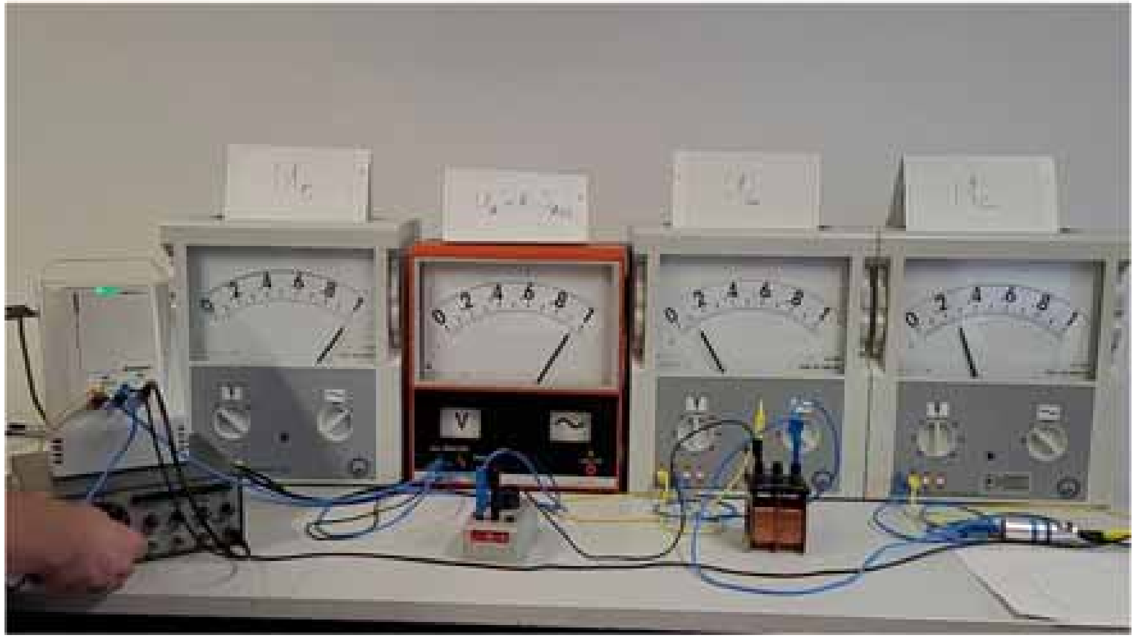
Halliday-Resnick-Walker (2003) Abb. 33.13

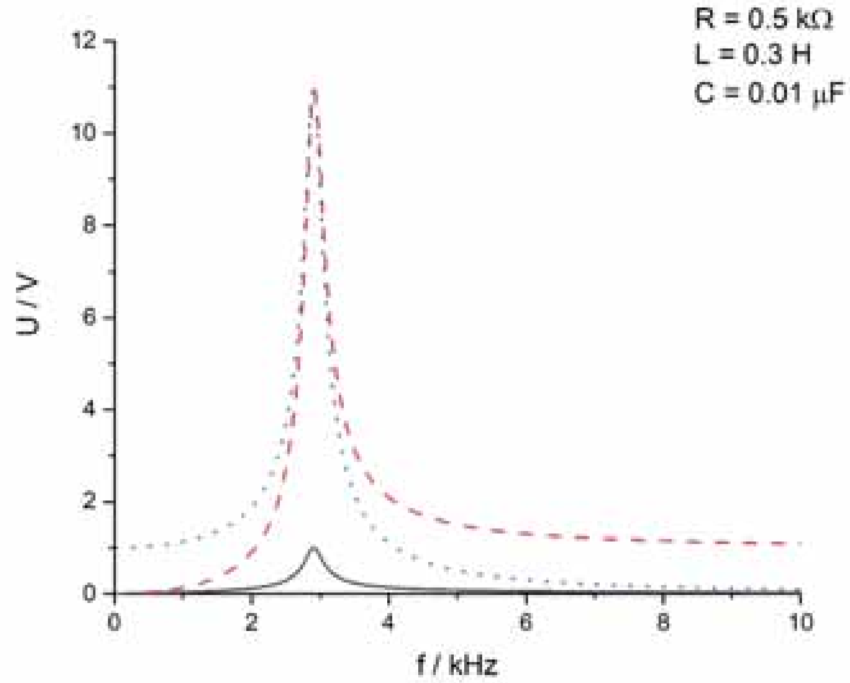
zu 2.7: Erzwungene Schwingung





zu 2.7: Erzwungene Schwingung





zu 2.8: Elektrische Leistung am Widerstand

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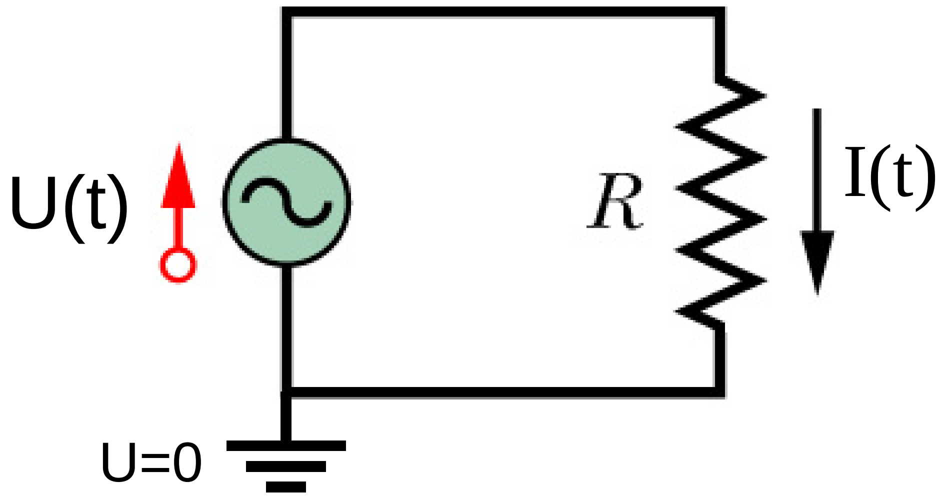
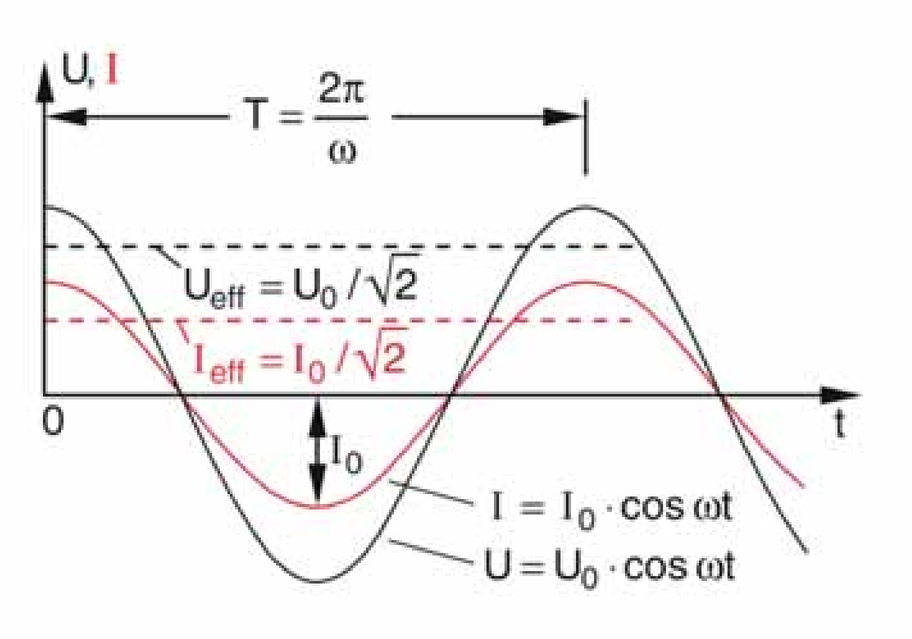
$$P_{el} = I_{eff}^2 \, R$$



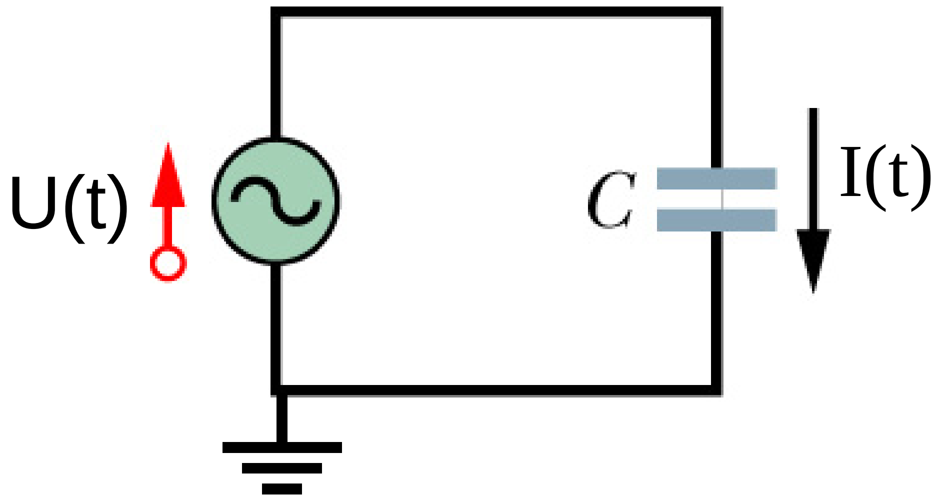
Halliday-Resnick-Walker (2003) Abb. 33.8



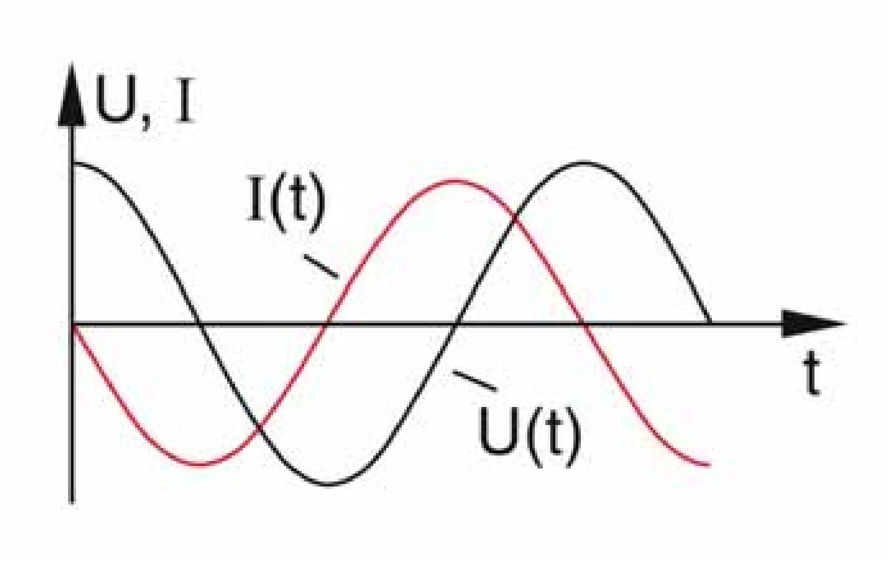
Demtröder – Experimentalphysik 2 (2013), Abb. 5.11.

zu 2.8: Elektrische Leistung am Kondensator

$$P_{el} = 0$$

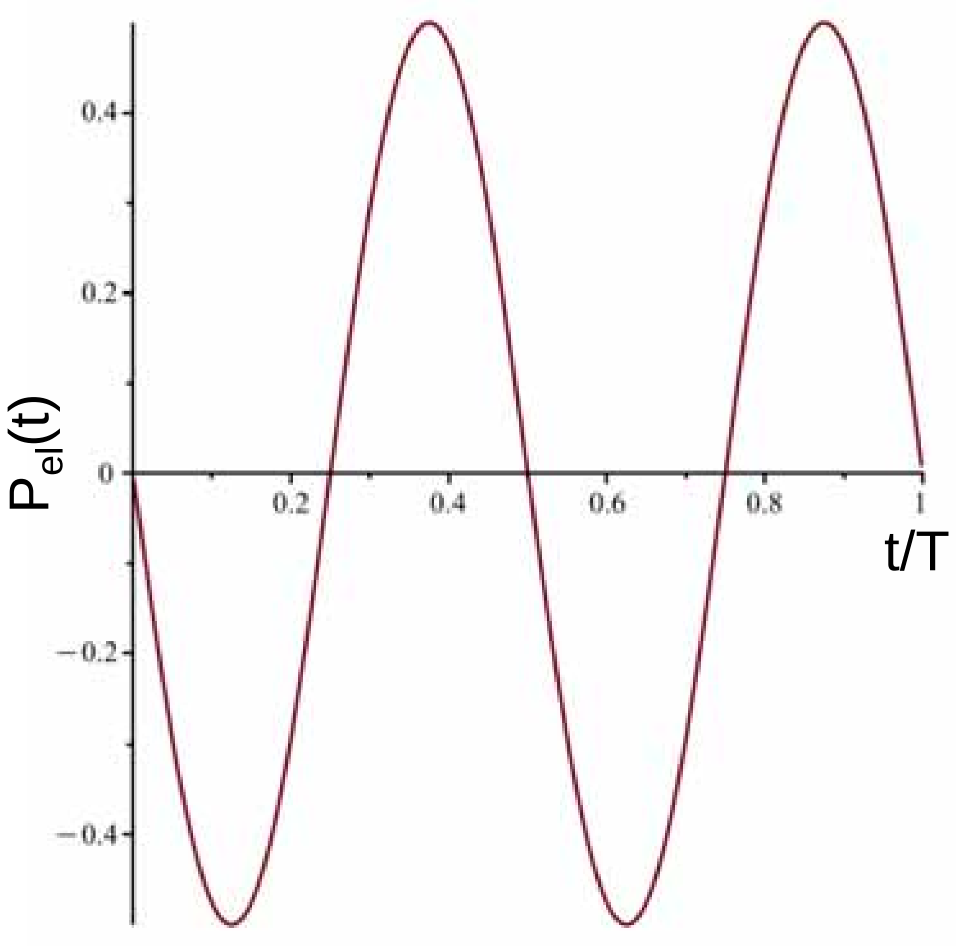


Halliday-Resnick-Walker (2003) Abb. 33.9



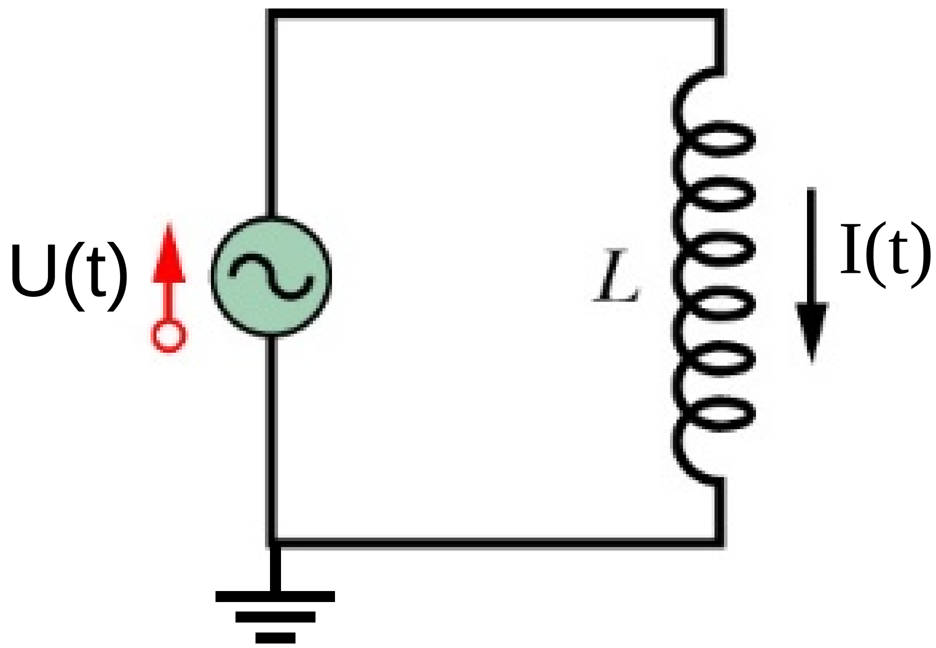
Demtröder – Experimentalphysik 2 (2013), Abb. 5.25.

zu 2.8: Elektrische Leistung am Kondensator

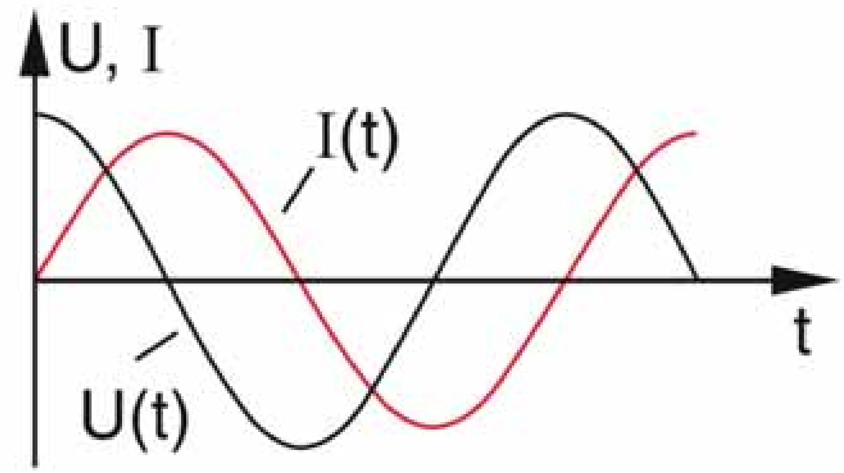


zu 2.8: Elektrische Leistung an der Spule

$$P_{el} = 0$$



Halliday-Resnick-Walker (2003) Abb. 33.9



Demtröder – Experimentalphysik 2 (2013), Abb. 5.23.